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Project Title – Swasthya Sathi – "Offline-First" Telemedicine System: Bridging the Healthcare Gap in Rural Nepal

Assignment 2 – Contextual Report

BSc (Hons) Computer Science and
Software Engineering

**CISO13-3 Research Methodologies
and Emerging Technologies**

University of Bedfordshire

Abstract

Swasthya Sathi is an innovative offline-first telemedicine platform designed to address the "oversight gap" in rural Nepal, where extreme geographical barriers, a severe shortage of healthcare workers (dropping to 1:150,000 in remote regions), and unstable internet connectivity render conventional digital health solutions ineffective. Focus is specifically placed on the continuous monitoring and management of Type 2 Diabetes Mellitus and Essential Hypertension, chronic conditions that require timely intervention to prevent acute complications.

The system is built upon four mutually reinforcing architectural pillars: a Local-First Architecture using SQLite and Hive for persistent on-device data storage; On-Device Edge Machine Learning (Logistic Regression/Decision Trees) via TensorFlow Lite for real-time risk classification with a target latency of <5ms; Tiered Asynchronous Communication that prioritises data minimisation and resilient delivery during low-bandwidth windows; and a triggered Offline First-Aid Protocol system providing WHO-standardised medical instructions immediately upon anomaly detection.

Primary research involving rural patients and medical professionals confirmed that network jitter and travel burdens—often exacerbated by gender-based socio-cultural constraints—are the primary obstacles to healthcare access. Technical evaluation of the platform focuses on maintaining a 97.5% classification accuracy for ML models and a 99% synchronisation success rate under adverse network conditions. By prioritising local intelligence over cloud dependency, Swasthya Sathi provides a scalable model for achieving healthcare equity in resource-constrained environments, facilitating measurable clinical improvements such as HbA1c reductions and enhanced blood pressure control regardless of a patient's connectivity status.

Acknowledgement

I would like to express my sincere gratitude to my academic supervisor and mentors at the university. Their expert guidance, technical insights, and consistent encouragement were instrumental in navigating the complexities of Edge ML and offline-first architecture throughout this project.

I also extend my thanks to the 13 patients from the districts of Sindhupalchok, Dhading, Rasuwa, Tanahu, and Solukhumbhu, as well as the 13 healthcare professionals who participated in the primary research. Their lived experiences and clinical expertise were vital in addressing the "oversight gap" in rural Nepal.

Finally, I am grateful to the Nepal Health Research Council (NHRC) and the World Health Organization (WHO) for providing the clinical benchmarks and first-aid protocols that ensure the system's medical reliability.



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Assessment 2: Contextual report

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Chapter 1: Introduction

1.1 Background

Access to healthcare services in Nepal is one of the most uneven across South Asia, being determined not only by its extreme geographical structure but also by the poor infrastructural development and the lack of sufficient numbers of healthcare workers. Telemedicine, which can be described as the arrangement of remote clinical services, via real-time two-way communication between the patient and the healthcare provider, using digital technologies like audio and visual means without the need for physical meeting between patient and the health care service provider. In spite of geographic barriers, telemedicine makes it possible to consult, monitor, and advise patients who are otherwise inaccessible due to their remoteness.

Despite its promising potential, the implementation of telemedicine services in Nepal faces an obstacle that is not well-appreciated which is internet connectivity. As more and more cities like Kathmandu, Pokhara, or Biratnagar become equipped with broadband networks, rural areas, where the majority of the 30 million Nepalese population lives, continue to experience irregular, slow, or even non-existent internet connectivity. While the number of doctors per patient nationwide is estimated at 1:3,000, in mountainous or hilly regions, this number may drop to 1:150,000.

This 'oversight gap' hits hardest to those patients who suffer from chronic non-communicable diseases (NCDs) like Type 2 Diabetes Mellitus and Essential Hypertension, who need constant monitoring and timely treatment in the case of acute situations, as well as constant modification of their treatment plans which cannot be done with technology that stops functioning when it loses access to the Internet. As a result, patients with health problems become most physically prone at times when help is least available.

Swasthya Sathi ("Offline-First" Telemedicine System) was designed specifically to counter this problem. Unlike other conventional online systems, which were adjusted for use without the Internet, Swasthya Sathi is envisioned to function first and foremost offline, using internet connection only as an additional tool, not as a core part of its structure.

1.2 Problem Statement

Currently existing telemedicine and digital health solutions in Nepal share a fundamental architectural flaw: they are designed for stable, high-bandwidth environments and degrade catastrophically or cease to function entirely when connectivity is unavailable. This dependency creates an 'oversight gap' that disproportionately affects rural patients managing chronic conditions such as diabetes (blood sugar dysregulation) and hypertension (elevated blood pressure), who require continuous clinical monitoring and timely intervention.

There are three serious compounding factors that define this problem. First, network unreliability: rural broadband penetration remains scarce, and available mobile data connections frequently suffer from high latency, jitter, and packet loss that renders real-time video consultation infeasible. Second, High Cost: even where connectivity nominally exists, the financial burden of sustained data usage is not viable for rural households operating near subsistence income. Third, the absence of immediate intervention mechanisms: even when a patient successfully transmits a reading indicating a health anomaly, the absence of synchronous doctor availability means that actionable guidance may arrive hours or days later safety window for acute events.

The result is a system that fails precisely the population it is intended to serve: chronically ill patients in low-connectivity environments, who face the compounded risk of unmonitored disease progression, delayed intervention, and an absence of first-response guidance during critical episodes.

1.3 Proposed Solution

Swasthya Sathi is an offline-first health monitoring platform engineered specifically for the technical and socio-economic realities of rural Nepal. The system is built around three mutually reinforcing architectural pillars:

Local-First Architecture: All data collection, storage, and clinical analysis occur on the patient's mobile device using SQLite or Hive for persistent local storage, ensuring that system functionality is entirely independent of network availability.

On-Device Edge Machine Learning: A lightweight Logistic Regression or Decision Tree model, deployed via TensorFlow Lite, performs real-time risk prediction and anomaly detection directly on the mobile device. This enables immediate clinical assessment without server round-trips, with target latency below 5 milliseconds.

Tiered Asynchronous Communication: Rather than depending on high-bandwidth video calls, the system employs a background synchronisation strategy that transmits critical alerts and health summaries to a remote doctor's web dashboard during any available low-bandwidth window — prioritising data minimisation and resilient delivery over real-time interaction.

Fourthly, there is another very important aspect that can be considered as an addition to the above-mentioned three pillars: a local database of first aid recommendations approved by WHO and specific to Nepal that would come up automatically in response to detecting a critical anomaly by the Edge ML algorithm.

1.4 Aim and Objectives

Aim: To research, design, and develop a secure, clinically functional, and offline-capable telemedicine platform featuring on-device Machine Learning, capable of bridging the patient-doctor oversight gap in rural Nepal.

The following objectives operationalise this aim:

- To conduct a systematic review of offline-first architectural patterns and edge computing frameworks applicable to resource-constrained healthcare environments.
- To develop a cross-platform mobile application using the Flutter SDK that supports full offline data collection, local storage, and on-device ML inference.
- To implement and optimise Edge ML models (Logistic Regression, Decision Tree) for real-time, low-latency risk prediction on entry-level Android smartphones.
- To engineer a robust asynchronous synchronisation mechanism capable of reconciling local and remote health records without data loss or conflict during intermittent connectivity.
- To design a triggered first-aid protocol system that automatically delivers WHO-standardised medical instructions to patients upon anomaly detection, without network dependency.
- To build a web-based doctor dashboard (React/Next.js) enabling remote medical professionals to review patient trends, analyse historical data, and adjust treatment plans asynchronously.
- To evaluate the system against clinical benchmarks for accuracy, security, usability, and socio-cultural impact, including gender-based accessibility barriers.

1.5 Scope

The scope of application for this clinical project is limited to the treatment of two chronic non-infectious diseases: type 2 diabetes mellitus (blood glucose level measurement) and essential hypertension (blood pressure measurement). These diseases have been chosen because of the frequency of occurrence in the countryside in Nepal, the crucial role played by continuous monitoring in treatment, and the serious potential danger of developing an acute complication in case monitoring stops.

The technical scope contains two primary artefacts: a Flutter-based mobile application targeting Android devices for patient-side use, and a React/Next.js web dashboard for doctor-side trend analysis and alert review. The system architecture assumes a zero-trust connectivity model, in which offline functionality is the guaranteed baseline and all network-dependent features (synchronisation, dashboard updates) are treated as opportunistic enhancements.

The project explicitly excludes real-time video consultation, wearable sensor integration, pharmaceutical prescription management, and emergency dispatch coordination in this iteration. These capabilities are noted as potential directions for future development in Chapter 8.

1.6 Intellectual Challenge

The primary intellectual challenge of this project lies at the intersection of three distinct technical domains. First, Available Bandwidth (ABW) estimation is notoriously difficult in mobile network environments characterised by jitter, variable delay, and dynamic spectrum allocation. Existing active estimation tools frequently exhibit limited accuracy and long convergence times, making it non-trivial to determine reliably when a synchronisation window is viable without incurring unacceptable data loss.

Second, the project must solve the data reconciliation problem inherent in any offline-first architecture: ensuring that records created or modified on the device during a connectivity outage are merged accurately with the authoritative server-side record when synchronisation eventually occurs. This requires conflict detection logic, timestamp management, and a robust SyncQueue implementation that can handle partial failures gracefully.

Third, optimizing Edge ML models for implementation on mobile hardware that is resource-constrained poses a substantial technical hurdle. The models need to be highly accurate in terms of their medical precision, which should be about 97.5 percent according to literature comparisons, but at the same time, maintain an inference latency rate of less than 5 milliseconds.

1.7 Rationale

The clinical reason behind Swasthya Sathi lies in the vast amount of research that has demonstrated the effectiveness of the digital health approach to managing chronic conditions. For example, researchers have found that participants experienced a decrease in their HbA1c levels by up to 2.9 points, while blood pressure changes were also significant in patients utilizing digital monitoring services. Yet, these results require constant monitoring – exactly what connectivity-based systems are unable to provide in the rural settings of Nepal.

From a socio-cultural perspective, telemedicine has proven itself effective in removing gender-based access limitations where local traditions limited the ability of women to travel to obtain treatment. Thus, this platform will address a very specific problem affecting a large portion of the Nepali rural community. What is more, the inclusion of Nepali clinical guidelines and local language increases the relevance and accessibility of this health care system for the targeted population group.

1.8 Contribution

Three primary contributions of Swasthya Sathi to the field of digital health in resource-limited environments can be identified. First, it presents a feasible architectural design of an offline-first telemedicine application that retains its clinical capability from zero to high network availability, adjusting dynamically, as opposed to graceful degradation. Second, it introduces an example of how to implement Edge ML solutions for the monitoring of chronic diseases without relying on cloud infrastructure and serves as a blueprint for such solutions in other poor and underserved regions around the world. Third, it explicitly tackles the issue of the ‘oversight gap’ using automated clinical protocols.

1.9 Structure of the Contextual Report

This report is organised into eight chapters that trace the project from conceptual foundation through to implementation planning and critical evaluation:

1. Introduction: Project background, problem definition, proposed solution, aims, objectives, scope, and rationale.
2. Project Plan: Detailed project timelines across two academic semesters, comprehensive risk analysis with mitigation strategies, and a conceptual workflow diagram.

3. Literature Review: Critical examination of telemedicine frameworks, the healthcare disparity in Nepal, offline-first architectural theory, and Edge ML applications in clinical settings.
4. Primary Research: Market analysis of existing telehealth products, survey design and methodology, and analysis of stakeholder data on user needs and technology acceptability.
5. Artefact Planning: Functional and non-functional requirements, software and hardware specifications, and system design artefacts including use case diagrams, database schema, and activity diagrams.
6. Critical Analysis and Evaluation: Testing strategy covering Edge ML validation, synchronisation testing, and security auditing; evaluation strategy covering clinical reliability, usability, socio-cultural impact, and digital literacy.
7. Implementation Plan: Phased development roadmap from environment setup through final delivery, mapped to the Semester 2 Gantt chart.
8. Conclusion: Summary of findings, assessment of project impact, and recommendations for future development.

Chapter 2: Project Plan

This chapter outlines the strategic planning framework underpinning the development of Swasthya Sathi. It encompasses a structured two-semester timeline with defined milestones, a systematic risk analysis with quantified impact assessments and mitigation strategies, and a conceptual workflow diagram illustrating the system's data pipeline from patient input to professional review.

2.1 Gantt Charts

The project is divided into two separate semesters that map respectively to research and planning activities, and technical implementation and validation. Each semester has its own Gantt chart reflecting task dependencies, durations, and progress tracking.

Semester 1: Research and Planning

Semester 1 focuses on establishing the intellectual and conceptual foundation for the project. The primary deliverables are a refined project proposal, a comprehensive literature review, and the completed Contextual Report (Assignment 2). All tasks in this semester are interconnected through a sequential finish-to-start dependency chain, ensuring that each phase builds directly on the evidence and conclusions of its predecessor.

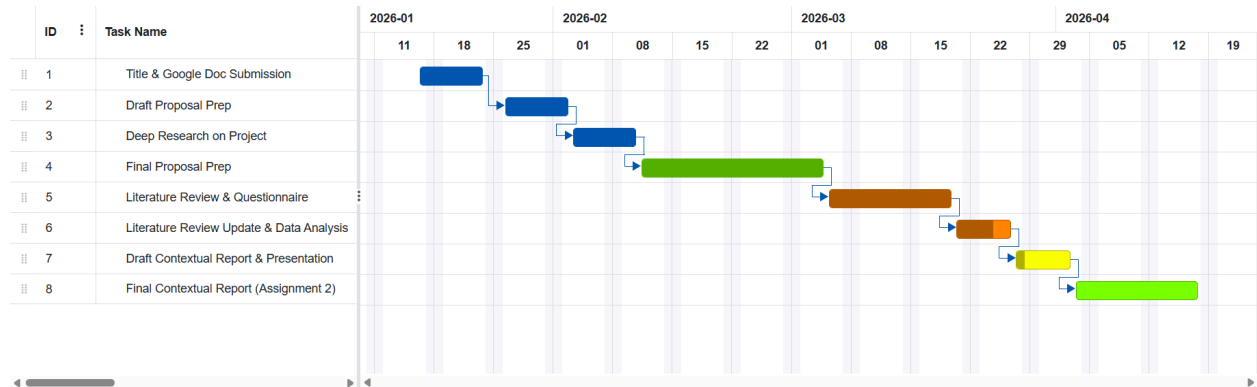


Fig 1: Gantt Charts of semester 1

Semester 2: Development and Implementation

Semester 2 transitions from research to technical execution. The development plan follows a modular, phase-based approach in which local-first patient functionality is built and secured before integration with the remote doctor dashboard and asynchronous synchronisation mechanisms. This sequencing ensures that the system's most critical offline capabilities are operational before more complex network-dependent features are introduced.

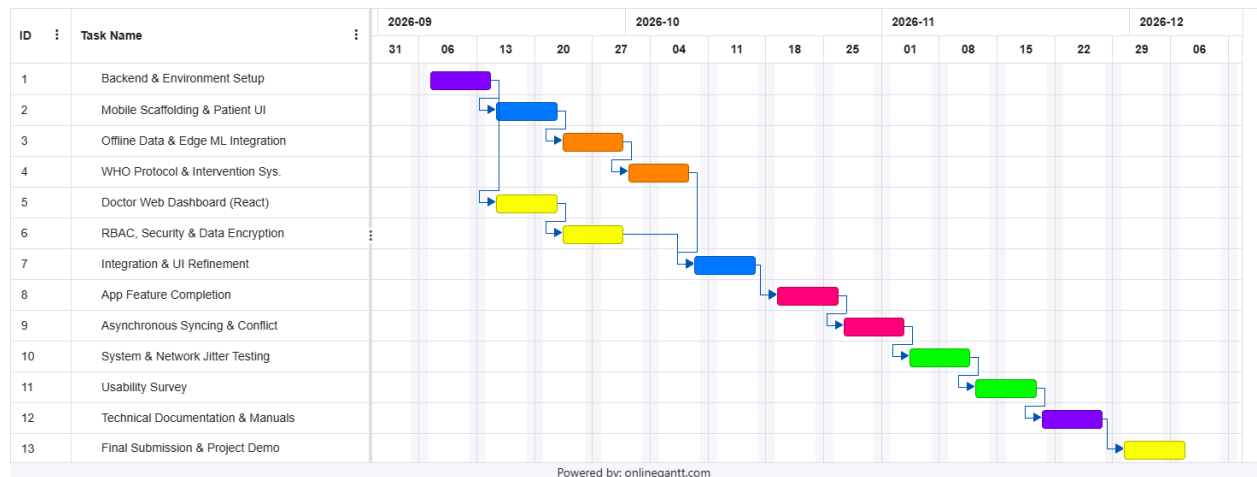


Fig 2: Gantt Charts of semester 2

2.2 Risk Analysis

A structured risk analysis has been conducted to identify threats to the successful delivery of Swasthya Sathi. Each risk is assessed against two dimensions — probability of occurrence and potential impact on the project — and a targeted mitigation strategy is assigned. All identified risks carry a High impact rating, reflecting the clinical context of the project in which system failures may have direct consequences for patient safety.

Risk ID	Possible Risk	Probability	Impact	Mitigation Approach
R1	Data Sync Conflicts	Medium	High	Implement robust database reconciliation and conflict resolution logic using timestamp-based versioning and the SyncQueue table. Test with simulated partial connectivity and concurrent writes.
R2	Edge ML Inaccuracy	Medium	High	Employ validated algorithms (Logistic Regression, Decision Tree) tested on representative local clinical datasets. Conduct cross-validation and subject-wise splits to ensure accuracy is not inflated by random-split methodology.
R3	Data Breaches / Unauthorised Access	Low	High	Implement AES-256 encryption for all locally stored data and enforce Role-Based Access Control (RBAC) on the doctor dashboard. Conduct penetration testing to verify the security posture.
R4	Time Management & Scope Creep	Medium	High	Adhere strictly to the Gantt chart milestones with weekly progress reviews. Scope is explicitly bounded (diabetes and hypertension only) to prevent feature drift during Semester 2 implementation.

R5	Incorrect First-Aid Guidance	Low	High	Source all first-aid protocols exclusively from WHO-approved clinical guidelines and Nepal Health Research Council recommendations. All protocols to be reviewed by a qualified medical professional prior to integration.
R6	Device Compatibility & Performance Constraints	Medium	Medium	Test on a range of entry-level Android devices with limited RAM (2GB) and processing power. Optimise TensorFlow Lite model size and inference pipeline to meet the <5ms latency target across device tiers.

2.3 Workflow Diagram

The conceptual workflow of Swasthya Sathi is organised as a six-stage pipeline, designed to bridge the oversight gap through a consistent local-first approach regardless of connectivity status.

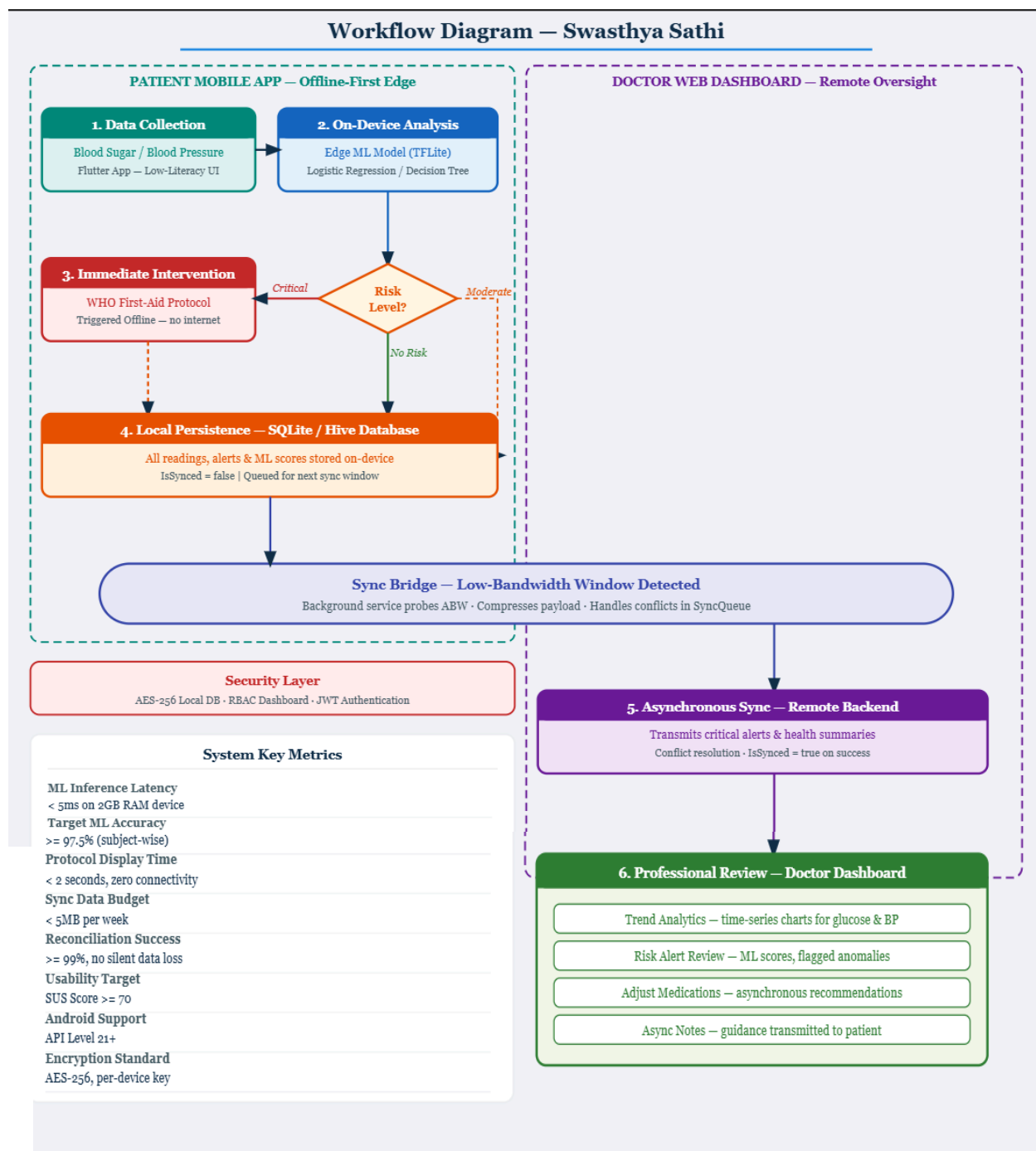


Fig 3: Workflow diagram of Sywasthya Sathi

These are the six Pipeline :

1. **Data Collection:** The patient enters blood glucose or blood pressure readings into the Flutter mobile application. The UI is designed for low digital literacy, with simplified data entry and clear visual feedback.
2. **On-Device Analysis:** The integrated Edge ML model (Logistic Regression or Decision Tree, deployed via TensorFlow Lite) processes the input locally to classify the reading as No Risk, Moderate Risk, or Critical Risk, generating an ML Prediction Score that is stored in the RiskAlerts table.
3. **Immediate Intervention:** If the ML model classifies the reading as Critical Risk, the application automatically retrieves and displays the relevant first-aid protocol from the local FirstAidProtocols table. These instructions are WHO-approved and displayed without any network dependency.
4. **Local Persistence:** All health readings, risk alerts, and protocol activations are stored in the local SQLite/Hive database. A sync status boolean in the HealthReadings table tracks which records have been transmitted to the server.
5. **Asynchronous Synchronisation:** A background service monitors available bandwidth. When a viable low-bandwidth window is detected, the application transmits pending records from the SyncQueue to the remote backend, with conflict resolution logic handling any discrepancies between local and server-side data.
6. **Professional Review:** The doctor accesses the web dashboard to review synchronised patient data, examine trend analytics, review flagged risk alerts, and asynchronously adjust medication recommendations or escalation plans based on historical patterns.

Chapter 3: Literature Review

3.1 Overview and Background

This literature review systematically examines the theoretical and empirical foundations underpinning Swasthya Sathi. It draws on peer-reviewed publications, technical standards documentation, and grey literature from global health organisations to evaluate four interconnected domains: the conceptual framework of telemedicine as a healthcare delivery paradigm; the structural healthcare disparity in Nepal that defines the deployment context; the architectural principles and technical constraints of offline-first mobile systems; and the application of Edge Machine Learning (Edge ML) in resource-constrained clinical environments.

The scope of this review is deliberately bound to literature that is directly applicable to the three architectural pillars of the proposed system: local-first data persistence, on-device clinical inference, and asynchronous synchronisation. Peer-reviewed sources were selected from journals indexed in Scopus and PubMed, supplemented by technical reports from the World Health Organisation (WHO) and Nepal Health Research Council (NHRC). The review period prioritises publications from 2017 onwards, with seminal foundational works included irrespective of publication date. Collectively, ten primary sources are reviewed, each mapped to a specific technical or clinical challenge within the Swasthya Sathi architecture.

A critical observation that emerges across the reviewed literature is the persistent tension between the clinical evidence base for telemedicine — which is broadly positive for chronic disease management (Saboo and Saboo, 2025; Suhlrie et al., 2025; Ngamvitroj, Pitthayapong and Dhippayom, 2025) — and the infrastructure assumptions that underpin most deployed systems. Standard telemedicine architectures presuppose stable, high-bandwidth connectivity; the rural Nepali context invalidates this assumption structurally. This gap between clinical efficacy and deployment feasibility is the central intellectual problem that Swasthya Sathi is designed to resolve.

3.2 Review of Existing Projects and Solutions

A critical survey of existing telemedicine deployments in Nepal and comparable low-income, low-connectivity settings reveals a consistent architectural vulnerability: near-universal dependency on high-bandwidth, video-centric connectivity. Platforms such as Nepal's public-private telehealth initiatives and internationally funded eHealth programmes have predominantly attempted to adapt online-first models to rural contexts, resulting in systems that function adequately in urban pilot environments but fail to scale meaningfully to the populations most in need.

The fundamental limitation of these approaches is the assumption of Available Bandwidth (ABW). Megyesi et al. (2017) document that existing active ABW estimation tools suffer from limited measurement accuracy and extended convergence times in mobile network environments characterised by high jitter and variable delay. This means that not only are the systems themselves bandwidth-dependent, but the mechanisms intended to help them adapt to poor connectivity are themselves unreliable under the precise conditions where they are most needed. This finding directly informs the decision within Swasthya Sathi to implement a custom lightweight bandwidth detection mechanism rather than relying on any existing ABW estimation library.

A secondary, and frequently underappreciated, failure mode identified across existing systems is insufficient attention to the human element of technology adoption. Paul et al. (2023) identify healthcare as a "soft target" for cyber-attacks and note that digital health programmes in similar contexts have frequently under-invested in digital literacy support, resulting in low adoption rates among populations with limited prior smartphone exposure. This finding has direct implications for the design approach adopted in Swasthya Sathi, which prioritises a simplified, low-cognitive-load user interface and includes a targeted digital literacy evaluation component in the Semester 2 evaluation strategy.

From a model-driven architecture perspective, Núñez et al. (2020) demonstrate that offline data persistence in native mobile applications can be engineered with significantly reduced development effort through meta-model-driven code generation, achieving modification speeds 2.85 times faster than manual coding. However, their study identifies a critical unresolved limitation: the absence of robust automatic data synchronisation and conflict resolution between local and remote databases. This gap directly corresponds to Risk R1 (Data Sync Conflicts) in the Swasthya Sathi risk register and underscores the need for a purpose-built SyncQueue implementation with timestamp-based conflict resolution logic.

Swasthya Sathi's competitive differentiation within the existing landscape is therefore clearly defined. It is the only solution in the identified market that combines offline-first operation, on-device Edge ML inference, automated protocol-triggered intervention, and asynchronous professional oversight in a single integrated platform designed specifically for the rural Nepali context.

3.3 Literature Review

3.3.1 Conceptual Framework of Telemedicine

Source: Bashshur, R.L. (1995)

The definition provided by the Foundational academy for telemedicine as the medical assistance delivered through telecommunication and computer technologies in place of face-to-face interaction between the doctor and patient has remained the same since its very beginning in the 1990s. Despite this, recent definitions have focused on four main aspects of any efficient telemedicine system: the presence of geographic distance between all participants of the process; the use of ICT as an instrument of communication; sufficient number of employees; and an organizational form that is suitable for remote delivery of health services.

The scientific community has demonstrated that when all four aspects of the efficient telemedicine service are present, telemedicine delivers outcomes similar to those of traditional treatment of chronic non-communicable diseases. Scientific papers confirm that patients diagnosed with type 2 diabetes receive telemedicine benefits such as decreasing their HbA1c levels by up to 2.9%, while people with hypertension can see changes in both systolic and diastolic blood pressure. Efficiency of telemedicine, however, highly depends on the level of patient compliance, which can hardly be guaranteed in rural Nepal, where connectivity problems exist.

3.3.2 Healthcare Disparity in Nepal: Structural Context

Source: Parajuli, R. and Doneys, P. (2017)

Nepal's healthcare system exhibits an intense form of rural-urban inequality in South Asia. Although the total doctor-patient ratio for Nepal is 1:3,000, which is already below the WHO recommendation of 1:1,000, the disparity becomes greater in rural and isolated areas where the local doctor-patient ratio might reach levels above 1:150,000. Under such circumstances, the scarcity of doctors is further exacerbated by the 'brain drain' phenomenon, where doctors congregate in urban settings because of better work conditions.

It is projected that nearly 80 percent of the population of Nepal resides in rural and semi-urban locations, where the problem of insufficient healthcare access is much more prevalent. Furthermore, the prevalence of non-communicable diseases like diabetes and hypertension has been increasing due to lifestyle and diet changes. Therefore, telemedicine programs seem structurally incapable of addressing the issue since they are unable to ensure proper connectivity between physicians and patients.

3.3.3 Offline-First Architecture: Theory and Application

Source: Megyesi, P., Botta, A., Aceto, G., Pescapé, A. and Molnár, S. (2017)

The idea of being offline first means that there is a fundamental change in assumptions about how the software works. While traditional assumptions about the internet in software development were based on the fact that the network is available, in an offline first approach the absence of the network becomes an expected operating mode, while its presence is used to improve the product's operation. The offline first system is the most optimal option for the use in a clinical setting in low connectivity areas due to the match between the system's failure modes and the user's needs. In a system dependent on the internet connection, loss of connectivity during a medical emergency will be seen as a failure, while in the offline first environment it will just be a part of regular operations without any effect on core functionality. Persistent database can be implemented using SQLite or Hive for their efficiency in different operating systems.

3.3.4 Edge Machine Learning in Clinical Settings

Source: Sreekrishnan, A., Lansberg, M., Singer, A.E., Tennant, R.S., Howard, M., Buncom, J., Castillo, M., Rajasekhar, V. and Kraler, L. (2024)

Edge ML is defined as implementing pre-trained machine learning models on users' smartphones, Internet-of-Things sensors, and other endpoint devices instead of on centralized cloud servers. From the point of view of clinical applications, Edge ML provides the possibility to perform inference instantly and without any connection to the server, which is a crucial factor in such applications when it is impossible to assume stable access to the network.

As far as machine learning models for predicting risks associated with chronic diseases are concerned, there is extensive literature on their efficiency. Logistic Regression and Decision Trees have proven to provide accurate predictions while categorizing patients based on risks associated with diabetes and hypertension with accuracies ranging from 85% to 97.5%. The most important aspect concerning this research is the fact that these two machine learning models do not demand much computing power and can be used to create a mobile application for Android at an entry-level smartphone using TensorFlow Lite.

3.3.5 Technological Constraints and the "Offline-First" Necessity

Source: Megyesi, P., Botta, A., Aceto, G., Pescapé, A. and Molnár, S. (2017)

A critical barrier to rural telemedicine is the management of Available Bandwidth (ABW). ABW is a dynamic characteristic representing the amount of traffic a path can support without affecting other flows, but it is notoriously difficult to estimate in traditional networks due to delay and jitter. Existing active estimation tools often suffer from limited accuracy and long convergence times.

These constraints necessitate a move toward "Offline-First" architectures. Traditional systems create an "oversight gap" when connectivity fails, leaving patients without guidance during health anomalies. By utilizing Edge ML for on-device anomaly detection and local storage (such as SQLite or Hive), a system can maintain functionality with zero connectivity. This allows for asynchronous communication, where data is collected and analyzed locally and synced only when a low-bandwidth window becomes available. Additionally, the integration of locally stored first-aid protocols provides immediate medical guidance during the unavoidable delays in doctor reviews.

3.3.6 The effectiveness of telemedicine in the prevention of type 2 diabetes mellitus: a systematic review and meta-analysis of interventions

Source: Suhlrie et al. ,2025

The systematic review and meta-analysis carried out by Suhlrie et al. (2025) is an essential empirical basis for the Swasthya Sathi initiative. This study analyzed the effectiveness of 52 telemedicine-based diet and physical activity interventions, providing evidence that technology-based healthcare programs have the capacity to effectively manage several metabolic risk factors of Type 2 Diabetes (T2D) in people who are either overweight or obese. The meta-analysis found statistically significant decreases in body mass (MD: -1.66 kg), body mass index (MD: -0.71 kg/m²), waist circumference (MD: -2.82 cm), and hemoglobin A1c levels (MD: -0.07%). Thus, telemedicine can be considered a clinically applicable technology-based approach for the prevention of T2D, which aligns with the Swasthya Sathi goal of addressing the "oversight gap" using technological support in distant communities.

Despite these favorable clinical results, however, it must be noted that the studies reveal an important "Digital Divide," underscoring that low-resource countries are inadequately represented and that interventions may encounter difficulties in sustaining

their effectiveness in the long term. The Swasthya Sathi program manages these concerns explicitly with its unique "offline-first" design that is built with a focus on operating in the high-risk, infrastructure-poor conditions of rural Nepal. Rather than following the more common digital health approaches where their effects are found to diminish in the post-intervention period, Swasthya Sathi places a premium on testing and long-term adherence.

It is crucial that the deployment of Edge ML with local data storage allows the operationalization of the findings by Suhlrie et al. (2025) in a difficult geographical situation. This enables the provision of "continuous medical guidance" required for effective prevention of T2D regardless of the lack of access to internet services. Through the localization of both predictive algorithms and patient records on the device, Swasthya Sathi ensures that the behavioral training necessary for sustaining the weight loss and HbA1c reduction discovered in the meta-analysis is achieved.

3.3.7 Self-Management in Post-Stroke Care

Source: Ngamvitroj, Pitthayapong and Dhippayom (2025)

The study by Ngamvitroj, Pitthayapong and Dhippayom (2025) provides evidence for the effectiveness of theory-based self-management interventions in improving blood pressure control among stroke survivors. Their meta-analysis of 18 trials (5,435 participants) showed a greater reduction in both systolic (MD -4.87 mmHg) and diastolic (MD -1.34 mmHg) blood pressure compared to control groups. Crucially, the authors found that the reduction in systolic blood pressure was significantly greater in trials lasting 6 months compared with those over 6 months, concluding that the effectiveness of these interventions tends to fade over time. This underscores the importance of the Swasthya Sathi architecture, which utilizes local storage and persistent Edge ML monitoring to sustain self-management behaviors. By moving away from intermittent, internet-dependent apps toward a persistent "offline-first" system, the project aims to solidify the role of self-management in secondary stroke prevention beyond the initial 6-month window.

3.3.8 CGM Integration in Pharmacist-Supported Care

Source: Lee et al. (2026)

The scoping review by Lee et al. (2026) maps the impact of Continuous Glucose Monitoring (CGM) within pharmacist-led diabetes services, finding that CGM-integrated care leads to significant HbA1c reductions ranging from -0.4% to -2.9%. Notably, these results outperformed CGM-integrated care that lacked pharmacist involvement, which only saw reductions between -0.5% and -0.8%. This suggests that technological monitoring, when paired with specialized oversight and data-driven medication titration, drastically enhances clinical outcomes.

This research is directly relevant to Swasthya Sathi, as it validates the effectiveness of digital health tools for chronic disease management. However, the study identifies a critical "Digital Divide" and a lack of evidence regarding humanistic and economic outcomes, as most research remains focused solely on clinical parameters. Swasthya Sathi addresses these gaps by leveraging its "offline-first" Edge ML architecture to ensure that high-tech monitoring remains accessible and sustainable in low-resource, rural settings where consistent internet access and specialist availability are limited.

3.3.9 Data Integrity and Resilience in Remote Chronic Disease Management

Source: Sharma, Agrawal and Kodamana (2022)

In their investigation on data reconciliation frameworks to address issues related to errors and inconsistencies in the dynamic operation of hybrid renewable energy systems (HRESs), Sharma et al. (2022) compared different methods: the Extended Kalman Filter (EKF), the Ensemble Kalman Filter (EnKF), and the Regularized Particle Filter (RPF). As a result of comparing the performance of these techniques, RPF turned out to be the most effective, having an estimation accuracy which was 75% higher than EKF and 50% higher than EnKF.

This work is highly relevant to the Swasthya Sathi project, which utilizes an "offline-first" architecture to bridge the healthcare gap in rural Nepal. Because the project faces risks associated with Data Sync Conflicts when reconciling local SQLite/Hive data with an online dashboard, the mathematical rigor of these reconciliation frameworks offers a professional blueprint for ensuring data integrity. The study adds value by validating a method for maintaining system reliability even during "sensor impairments," which is directly analogous to the intermittent connectivity and data corruption challenges the system must navigate.

However, a key area for improvement is the computational trade-off, as high-accuracy filters like RPF require more processing power than simpler variants. For Swasthya Sathi, future work must focus on optimizing these reconciliation logic sets for Edge ML environments with limited hardware resources to ensure that real-time anomaly detection remains efficient despite only a

3.3.10 Model-Driven Architecture for Offline Data Persistence

Source: Núñez et al. (2020)

The research article by Núñez et al. (2020) introduces MoWebA Mobile, a model-driven development (MDD) framework designed to streamline native mobile application creation by focusing specifically on the data layer. The authors implemented a system of meta-models and transformation rules that automatically generate code for platforms like Android, specifically targeting local data persistence (SQLite, files, and key-value pairs) to handle network instability. Their findings indicate that this approach significantly improves portability and reduces development effort, allowing modifications to be made 2.85 times faster than manual coding.

This research is fundamentally relevant to the Swasthya Sathi project, which utilizes an "offline-first" architecture with SQLite and Hive for chronic disease management in rural Nepal. It adds professional value by providing a rigorous architectural blueprint for asynchronous data handling, validating your strategy of utilizing on-device storage to maintain medical monitoring during zero-connectivity periods. However, the study identifies a critical area for improvement: the current limitation in automatic data synchronization and conflict resolution between local and remote servers. This directly underscores your project's need to prioritize robust reconciliation logic to mitigate Data Sync Conflicts (Risk R1), ensuring that health anomalies detected offline are accurately transmitted to doctors once a low-bandwidth window permits.

3.3.11 Edge-Computing for Real-Time Cardiac Monitoring

Source: Utsha and Morshed (2025)

The research by Utsha and Morshed (2025) introduces a GAN-augmented hybrid CNN-LSTM model for real-time cardiac arrhythmia detection via the CardioHelp smartphone app. To address severe class imbalance in ECG datasets, the authors implemented Generative Adversarial Networks (GANs) to synthesize rare arrhythmic beats, which are then processed through a hybrid architecture designed for spatial and temporal feature extraction. Their findings demonstrate a robust validation accuracy of 97.50% when deployed on edge devices, with an exceptionally low average inference time of only 4.6 ms.

The current research has tremendous implications for the Swasthya Sathi program since it demonstrates that there is an ability to use complex deep learning algorithms on constrained mobile devices in order to provide the "offline-first" model of health care service delivery. The results can serve as a benchmark for the professional implementation of such algorithms in terms of providing low latency rates and efficient CPU performance levels (below 20%).

Nonetheless, there is room for improvement, which is also important to take into account in relation to the Swasthya Sathi program. In particular, the fact that healthy participants were used for the experiment, as well as random split validation of the dataset, might lead to the overestimation of accuracy levels, which would not reflect the clinical accuracy level achieved in working with real-world patients' data. Therefore, further efforts should be made to validate the algorithm in terms of each participant individually and to examine the influence of user movement and noise.

3.3.12 Privacy and Security in Digitized Healthcare

Source: Paul et al. (2023)

The research article by Paul et al. (2023) examines the critical privacy and security challenges emerging from the digitization of healthcare, specifically focusing on the Internet of Medical Things (IoMT) and Electronic Health Records (EHR). The authors identify the healthcare sector as a "soft target" for cyber-attacks, noting that ransomware and data breaches in this industry result in average recovery costs reaching \$7.1 million—over three times the cost of incidents in other sectors. To combat these mounting threats, the study evaluates advanced security

paradigms, including blockchain for decentralized data integrity and federated learning for privacy-preserving analysis.

This paper is highly relevant to the Swasthya Sathi initiative since it tackles the problem of mitigating Risk R3 (Data Breaches) in relation to the "offline-first" architecture you are using. Indeed, this study gives a strong technological basis that proves the adequacy of your approach, namely the usage of local database encryption (AES) and Role-Based Access Control (RBAC), which makes data breaches much less probable because they reduce the risk of endpoints' leaking data. By the very principle of your offline-first technology, any risks related to being attacked through the Internet are automatically reduced since your data is stored locally.

Nevertheless, as noted in the paper, it is important not to rely only on technical security measures, as many health management software solutions lack optimization. It is also necessary to pay more attention to the human factor. In order to do this successfully, it would be essential for the Swasthya Sathi team to involve patients actively and teach them how to use their new technological environment securely.

3.3.13 Edge Machine Learning for AI-Enabled IoT Devices (Merenda et al., 2020)

Source: Merenda et al. (2020)

Switching from cloud-dependent to edge processing is crucial for enabling AI in IoT, especially where resources are scarce. As explained by Merenda et al. (2020), optimization strategies such as pruning and quantization allow "conscious things" to conduct local anomaly detections and make real-time decisions independent of internet connectivity. The "offline-first" strategy will be vital for the success of the Swasthya Sathi initiative in rural Nepal since it ensures consistent patient monitoring and automatic first-aid without depending on any internet connections.

Apart from that, performing calculations locally makes it easier for us to protect patients' sensitive medical records from cyber-attacks by limiting their communication. Additionally, this development contributes to saving energy as the devices no longer have to send large data amounts across the network. This innovation opens opportunities for implementing asynchronous synchronization methods that can connect patients living in rural areas to urban medical centers at an incredible accuracy rate.

3.3.14 Logistic Regression (LaValley, 2008)

Source: LaValley, 2008

This research provides the statistical and clinical blueprint for the Swasthya Sathi risk prediction models. Since the project targets binary health outcomes (e.g., "Critical" vs. "Stable" readings for blood sugar or pressure), logistic regression offers a mathematically robust method to process these dichotomous results using continuous predictors. The paper establishes a vital "rule of thumb" for model stability, requiring at least 10 events per predictor to ensure reliable estimates—a guideline that directly informs the training of the project's Edge ML models. By adhering to the goodness-of-fit assessments and odds-ratio methodologies outlined in this literature, the system can provide clinically reliable diagnostics comparable to established cardiovascular studies like the Framingham Heart Study.

3.3.15: A Review on Decision Tree Algorithm in Healthcare Applications

Source: Abdulqader & Abdulazeez, 2024

The research paper titled "A Review on Decision Tree Algorithm in Healthcare Applications" highlights the pivotal role of decision tree algorithms in enhancing medical diagnostics, prognosis, and health monitoring. These algorithms are particularly valued for their interpretative and transparent nature, utilizing a tree-like structure to partition complex datasets into simpler, actionable decision outcomes. This transparency is essential in clinical settings, as it allows medical practitioners to clearly understand and explain the logic behind automated decisions, thereby fostering patient trust and clinical acceptance.

The results of technical evaluations in the paper reveal that decision tree techniques are very efficient in detecting chronic diseases. They tend to exceed other approaches, such as SVM and Naive Bayes, in terms of efficiency and training. For example, some implementations provided an accuracy rate of 96.1% in predicting the development of heart disease but required much less time for training. In addition, the paper reviews the application of decision trees to IoT devices, which enables them to diagnose health problems in real time without disclosing personal data from patients during cyberattacks.

The research paper is crucial for the Swasthya Sathi initiative due to its relevance to implementing Edge ML in chronic disease detection in rural Nepal. The validation of decision tree techniques in predicting diabetes and hypertension is essential for achieving the clinical aim of overcoming the "oversight gap". As decision trees require

minimal computing power and allow making decisions in real time during home health monitoring, they serve as a powerful rationale for the project implementation.

3.3.16: Artificial Intelligence Algorithms for Healthcare

Source:Chumachenko & Yakovlev, 2024

The editorial article named "Artificial Intelligence Algorithms for Healthcare" brings out how the integration of AI can transform the health care system through proper diagnosis, successful prediction of outcomes, and tailored treatment plans. According to the research, incorporating AI in the health care industry is currently happening, and the technology is changing the processes involved by analyzing big data in search for patterns that could not have been recognized by conventional means. Apart from its benefits, the study identifies some concerns associated with the use of AI like privacy of data, bias in algorithms, and need for explainability of AI.

The proposed study bears significant relevance to the Swasthya Sathi project in that it supports the clinical and technological transition to healthcare systems driven by technology in marginalized areas. In particular, the analysis of methods for timely detection of diseases with hardware-assisted machine learning models is an excellent theoretical analogy for the application of Edge ML in the prediction of risks in limited environments. Additionally, the consideration of ethical data collection and protection in the paper justifies the security measures taken within the framework of the project, namely the application of AES encryption and Role-Based Access Control (RBAC). As such, the project utilizes artificial intelligence to overcome the "oversight gap" effectively.

3.3.17: Telemedicine Practice: Review of the Current Ethical and Legal Challenges

Source: Nittari et al., 2020

From the research paper entitled "Telemedicine Practice: Review of the Current Ethical and Legal Challenges," it is apparent that although telemedicine proves extremely efficient in treating patients located in far off or disadvantaged areas, its universal use is often impeded due to difficult issues related to ethics and laws. Among the main problems raised in the study include the non-existence of uniform laws concerning negligence and accountability, inconsistency of consent policies from one country to another, and the increased risk to privacy linked to electronic patient records. From the findings presented, it is evident that for telemedicine to be fully embraced, the standardization of laws and health training for telenegligence should occur.

The current study has major relevance to the Swasthya Sathi initiative since it creates a professional basis for the importance given by the project team to issues of security and confidentiality in their approach to patient data management. First, the authors' recognition that the healthcare industry makes a good "soft target" for attacks by cyber-criminals fully justifies Swasthya Sathi's choice of AES encryption and role-based access control as protective measures for rural data kept in local databases. Second, attention drawn in the paper to the "human factor" highlights the validity of combining technology and digital literacy training in the proposed project to gain the trust of rural residents in Nepal.

3.3.18: The Current Status of Telemedicine Technology Use Across the World Health Organization European Region

Source: Saigí-Rubió et al., 2022

The research paper titled "The Current Status of Telemedicine Technology Use Across the World Health Organization European Region" provides a comprehensive overview of 33 systematic reviews encompassing over 2,239 primary studies. The results demonstrate a clear clinical benefit of telemedicine in the screening, diagnosis, and long-term management of chronic conditions, including cardiovascular and respiratory diseases. However, the research identifies significant barriers to implementation, specifically within the individual domain (shortcomings in technology-related knowledge and skills) and the technological domain (persistent issues with internet access and infrastructure). It concludes that while telemedicine significantly enhances clinical outcomes and patient satisfaction, its generalized adoption depends on addressing these user-related barriers and establishing standardized legal and ethical frameworks.

The research holds immense relevance to the Swasthya Sathi project because it validates, on a broad scale, the application of digital health technology in the management of chronic conditions like diabetes and hypertension that fall under the purview of the project's scope. The challenges raised concerning internet connectivity and infrastructure validate the need for the offline first approach of the project, in addition to the use of edge ml technology, which ensures clinical capability without internet connectivity. The research also highlights the challenges posed within the individual domain related to a lack of digital proficiency, thus validating the project's approach to ensuring digital literacy.

Swasthya Sathi is able to translate the lessons from global literature by adopting these facilitators, such as patient empowerment and response time, to create a solution that not only works but is highly resilient. The focus on data privacy in the paper further reinforces the project's need to adopt AES encryption and Role-Based Access Control. Finally, the study serves to support the project objective by indicating that, with an appropriately developed telemedicine system, the "oversight gap" can be effectively closed, leading to better patient follow-ups and outcomes.

3.4 Summary of Literature Review: Findings and Critical Analysis

According to the literature and findings from the primary surveys, although telemedicine has been proved effective for the management of chronic conditions, its efficiency in rural areas in Nepal is affected due to the existing "oversight gap" because of the presence of several infrastructural and social barriers. The recent benchmarks in the use of technology in the health care industry show that through digital monitoring, there can be a reduction of up to 2.9% in HbA1c levels along with a substantial reduction in the level of blood pressure. However, the primary data collected from five rural regions, including Sindhupalchok and Solukhumbhu, shows that these benefits are inhibited because of the presence of up to 24 hours of travel time and network issues.

Also, the deployment process will have to make difficult socio-technical compromises in order to ensure the safety of the patient and accessibility. Technically speaking, this project is based on subject-wise cross-validation, which does not allow for the overestimation of accuracy as seen in current CNN-LSTM models, and combines the requirement for high-quality data matching with the processing capacity constraints imposed by the Android phone. Additionally, this project is designed with consideration for the dual goals of enhancing digital literacy and gender equality, as the telemedicine technology acts as an essential means for women in Nepal, who are burdened with household chores and unable to travel on their own to seek medical care.

3.5 Justification

The offline-first, Edge ML architecture of Swasthya Sathi is comprehensively justified by the convergent evidence from the literature. No existing solution identified in the market

analysis provides persistent, offline-capable clinical monitoring with immediate automated intervention and asynchronous professional oversight in a single integrated platform targeted at the specific constraints of rural Nepal. The research gap is both genuine and consequential: the populations most at risk from chronic non-communicable disease in Nepal are precisely those for whom current telemedicine solutions are least effective, due to the fundamental architectural mismatch between online-first system design and the offline-first deployment reality.

The clinical justification is grounded in the well-documented efficacy of structured digital monitoring for Type 2 diabetes (Saboo and Saboo, 2025; Suhlrie et al., 2025; Lee et al., 2026) and hypertension management (Sreekrishnan et al., 2024; Ngamvitroj, Pitthayapong and Dhippayom, 2025). The adoption of Logistic Regression and Decision Tree models via TensorFlow Lite is validated by comparable edge deployment benchmarks in the literature, including the 97.50% accuracy and sub-5ms inference latency achieved by Utsha and Morshed (2025) in a cardiac monitoring edge application. The architectural approach to local data persistence via SQLite/Hive is validated by Núñez et al. (2020), whose model-driven framework demonstrates the viability of offline-first native mobile applications at reduced development cost, and by the data reconciliation frameworks evaluated by Sharma, Agrawal and Kodamana (2022), which provide a mathematical foundation for ensuring data integrity across the asynchronous synchronisation cycle.

The security system, that is AES-256 encryption along with the role-based access control, is validated based on the work by Paul et al. (2023), where the authors argue the vulnerability of the healthcare industry to attacks on information infrastructure and mention average financial losses from such an attack equaling over \$7.1 million. Socio-cultural validation for the platform's features, including the user-friendly interface, confidential health-related communication channels, and gender-sensitive design, comes from Parajuli & Doneys (2017) – their study of telemedicine application in remote areas of Nepal shows a decrease in gender-related constraints on accessing healthcare services.

The primary data gathered during the Healthcare Access and Telemedicine Acceptability Study and Clinical Efficacy and Technical Constraints survey offers empirical support to both of these justifications mentioned above. Patients confirm that conventional telemedicine is beyond their reach because of long distances, household chores, and lack of technical knowledge. Clinicians state that network jitter and delays associated with reviews prevent them from effective monitoring of the patients' health status. Overall, combining the literature and collected data enables building a multi-faceted case regarding the clinical necessity and technical feasibility of Swasthya Sathi.

3.5 Technical Research

The technical foundation of Swasthya Sathi is designed to uphold a zero-trust connectivity model, where offline functionality is the guaranteed baseline and network-dependent features are treated as opportunistic enhancements. The following research details the selection and justification for the project's technical stack.

3.5.1. Frontend

The frontend is split into two distinct platforms to serve the unique needs of rural patients and urban doctors.

Patient-Side (Mobile): The Flutter SDK (Dart) was selected for the mobile application. Flutter allows for a single codebase that delivers native performance, which is critical for executing Edge ML inference via TensorFlow Lite with a target latency of under 5 milliseconds. Furthermore, Flutter's rich ecosystem supports "local-first" data handling, ensuring the app remains fully interactive even in airplane mode. The UI is specifically researched to address the digital literacy gap found in rural Nepal; it prioritizes large touch targets, meaningful iconography, and minimal text entry to ensure usability for users with limited smartphone experience.

Doctor-Side (Web): The professional dashboard is built using React and Next.js. This stack was chosen for its component-based architecture and its ability to handle complex data visualizations, such as longitudinal trend analytics for blood glucose and blood pressure. It also facilitates the implementation of Role-Based Access Control (RBAC) to ensure doctors only access data for their assigned patients.

3.5.2. Backend

The backend serves as the bridge for data synchronization and professional oversight.

API and Framework: The system utilizes Node.js with Express (or alternatively Firebase) to provide a lightweight REST API for synchronization endpoints. This choice prioritizes efficiency, as the synchronization payload must be minimized (targeting less than 5MB per week) to remain cost-effective for rural households.

Authentication and Security: To protect sensitive clinical data, the backend employs JSON Web Tokens (JWT) for stateless authentication. This is paired with RBAC middleware on all protected endpoints to prevent unauthorized access or data breaches.

3.5.3. Database

Swasthya Sathi utilizes a local-first persistence strategy to ensure that medical monitoring never stops during connectivity outages.

Local Storage (SQLite & Hive): On-device data is managed through a hybrid approach. SQLite provides relational data integrity for complex tables like the SyncQueue, while Hive offers high-performance NoSQL storage for Flutter-native operations. This local layer stores everything from health readings to WHO-approved first-aid protocols.

Data Protection: Every record stored locally is protected by AES-256 encryption. This ensures that even if a device is physically compromised, the patient's medical history remains unreadable.

Data Reconciliation: The remote database acts as the authoritative record. A custom SyncQueue table handles the operational logic for conflict resolution, using a timestamp-based "last-write-wins" policy while flagging critical risk conflicts for manual review.

3.5.4. Communication Strategy

Traditional real-time communication protocols like Web Sockets were researched but ultimately rejected as the primary synchronization method for rural Nepal.

The Problem with Real-Time: Web Sockets require a persistent, stable connection. However, rural Nepal is characterized by unstable 3G networks with high jitter, variable delay, and significant packet loss. These conditions cause real-time protocols to degrade catastrophically.

The Asynchronous Solution: Swasthya Sathi replaces persistent sockets with Tiered Asynchronous Communication. A background service performs bandwidth probing to identify viable low-bandwidth windows. Once a window is detected, it transmits pending records from the SyncQueue. This "Sync Bridge" ensures that data eventually reaches the doctor without requiring the patient to wait for a live connection to use the app.

Chapter 4: Market Research

A structured market analysis was conducted to characterise the current landscape of telemedicine solutions operating in or adapted for the Nepali market. The analysis examined products across three categories: general-purpose teleconsultation platforms, government and locally developed digital health initiatives, and academic or NGO-funded pilot programmes. Each category was assessed against the architectural criteria most relevant to Swasthya Sathi's target deployment environment — specifically, offline capability, on-device clinical intelligence, chronic disease monitoring support, and suitability for patients with limited digital literacy.

4.1 Market Context and Scale

Nepal's digital health market is expanding rapidly, with revenue projected to reach approximately USD 52.65 million by 2025. While internet penetration reached 55.8% of the population as of January 2025—with mobile devices accounting for 96% of that access—these figures mask a significant structural digital divide. Urban areas like the Kathmandu Valley enjoy connectivity rates of nearly 80%, whereas rural households average only 17.4%, dropping as low as 14% in remote provinces like Karnali. Furthermore, while 4G coverage is technically broad, rural users still rely on unstable 3G networks characterized by high jitter and variable delay, which makes the high-bandwidth video consultations used by standard telemedicine platforms structurally infeasible for most. (Himalayan Times, 2024; Megyesi et al., 2017).

4.2 Market Research (Existing Products)

The current telemedicine landscape in Nepal is categorized into commercial platforms, government initiatives, and NGO-funded pilots. Commercial apps such as Hamro Doctor, Cura Health, and Health Yaad Aayo offer synchronous video and audio consultations but are largely restricted to urban centers where broadband is stable. Government efforts include the National Teleconsultation Centre at Bir Hospital, which provides voice-based advice via a toll-free number, and provincial hub-and-spoke models designed to link rural health workers with urban specialists. Additionally, NGO and academic projects, such as the Internet Society's wireless system in Dullu, have demonstrated the clinical value of remote monitoring but remain heavily dependent on external funding and specific local infrastructure, limiting their long-term scalability across Nepal's difficult terrain.

4.3 Market Limitation

A critical analysis identifies five structural limitations within the existing market. First, there is a universal connectivity dependency, as all identified startups and government initiatives require live internet for core functionality. Second, most platforms focus on episodic consultations and lack the longitudinal monitoring capabilities necessary for chronic conditions like diabetes and hypertension. Third, current systems lack on-device clinical intelligence, routing all judgment

through live doctors, which is unsustainable given the extreme 1:150,000 doctor-to-patient ratio in remote districts. Fourth, there is an absence of automated intervention mechanisms to provide immediate first-aid guidance during clinical anomalies. Finally, many initiatives suffer from operational and financial fragility, often facing interruptions due to connectivity failures or the cessation of time-limited donor funding.

4.4 Market Need

The clinical need for an alternative solution is grounded in Nepal's significant burden of non-communicable diseases (NCDs), with diabetes prevalence at 8.5–10% and hypertension affecting roughly 21% of the rural population. There is a massive unmet demand for monitoring among the 23 million citizens living in rural or semi-urban areas who are currently excluded from video-centric urban platforms. The market requires a platform that can function in zero-connectivity environments to provide data collection and risk assessment entirely offline. Furthermore, there is a need for tools that address digital literacy gaps and gender-based barriers to access, providing a private and accessible environment for users who may have limited experience with technology or difficulty traveling for in-person consultations

Chapter 5: Primary Research

This chapter outlines the primary research carried out to prove the necessity of Swasthya Sathi and to collect the necessary information from various stakeholders regarding technological limitations, user requirements, and acceptance criteria that should guide the design of the system. Primary research is vital in this study because only secondary sources can help to set up the background of the clinical and technical knowledge, but they do not account for the actual conditions of the implementation environment. Two sets of surveys were created and implemented; one focused on the patients' end, and another surveyed medical practitioners with experience in managing patients remotely.

5.1 Objective of Primary Research

The general purpose of the main stage of the study was, in addition to confirming the already proven in scholarly works structural diagnosis of the issue, to collect empirical evidence concerning the practicality of that issue as experienced by people who will be using the Swasthya Sathi solution. The research sought to:

- Determine and measure the technological barriers, especially those related to limited access to bandwidth, devices and high costs associated with data usage which make the implementation of conventional telemedicine impossible.

- Determine the level of readiness of both patients and clinicians to implement an offline-first healthcare monitoring solution as well as their attitude towards ML-powered alerts and intervention triggers.
- Determine the level of digital literacy in the patient community and the usability needs the application interface should satisfy.
- Determine the trust levels associated with digitisation of health data and the types of information security measures that could gain the trust of all key stakeholders taking into account the fact that healthcare systems are known as "soft targets" for cyber-attacks (Paul et al., 2023).
- Collect quantitative and qualitative data on current health management practices – including how patients currently manage chronic diseases in the absence of digitisation tools.

5.2 Survey Questionnaire Design

Collection of primary data was carried out using two structured questionnaires conducted simultaneously. One questionnaire, entitled "Study on Access to Healthcare and Acceptability of Telemedicine Services - Swasthya Sathi," was meant for patients' side respondents from Nepalese rural areas. Another questionnaire, named "Evaluation of Clinical Success and Technical Challenges in Rural Telemedicine," was meant for healthcare professionals familiar with telemedicine services. Both questionnaires had been developed in accordance with the ethical guidelines for the BSc (Hons) Software Engineering project.

These questions were organized into four different domains:

- Access and Connectivity: These focused on travel time to health centers, gender barriers to independently traveling for medical assistance, device ownership, and app use comfort.

- **Household/Socio-Cultural Hindrances:** These questions revolved around how often household duties hindered participants from accessing medical treatment, as well as the privacy advantages of using an app over an actual consultation.
- **Features Acceptance:** Questions related to participant responses to certain features proposed in the system design based on various scenarios where a doctor's unavailability was due to poor internet connectivity, such as on-device risk calculation and automatic instructions for applying first aid.
- **Demographic Data:** Age, gender, district, and educational status data was also asked from patients in order to conduct a demographic analysis.

The clinical professional questionnaire was structured around four corresponding domains:

- **Constraints of Patient Load and Supervision:** Questions measured the patient load in rural regions per month, along with the lag time from patient anomaly detection to expert examination, due to network constraints.
- **Evidence for Clinical Monitoring:** Questions considered the professional opinion on the importance of six months of monitoring to achieve positive BP and HbA1c changes, and the importance of a remote monitoring framework, like MMAS-8, to ensure compliance.
- **Effect of Technical Constraints:** Questions considered the extent to which network jitter affected the consistency of patient health trends available to the reviewing professional.
- **Acceptance of Edge ML:** Questions considered professional acceptance of on-device artificial intelligence for detecting hypertension and hyperglycaemia alerts on a five-point Likert scale.

5.3 Data Collection and Analysis

Data collection was conducted between 5 March and 19 March 2026, followed by an analysis phase running from 20 to 26 March 2026. Digital survey forms were used to maximise reach across a demographically diverse participant pool. Thirteen patient-side responses were collected from participants across five rural districts: Sindhupalchok, Dhading, Rasuwa, Tanahu, and Solukhumbu. Thirteen clinical professional responses

were also collected. All responses were submitted anonymously, with informed consent mechanisms embedded directly in the survey instrument.

5.3.1 Patient Survey Findings

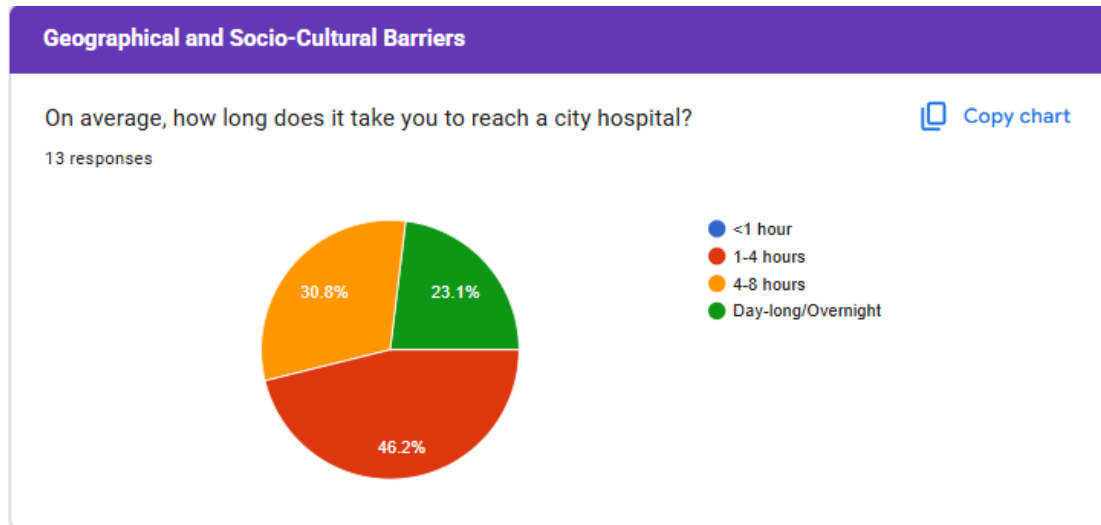


Fig 4: Geographical and Socio-Cultural Barriers Survey Questionaries

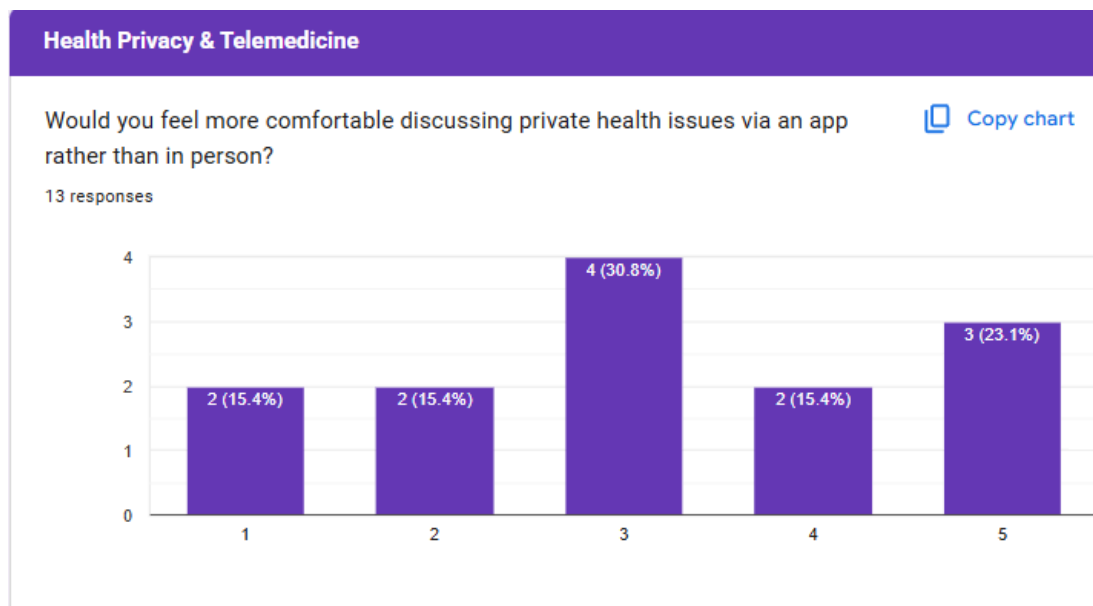


Fig 5: Health Privacy & telemedicine Survey Questionaries

If a doctor is unavailable due to poor internet, would you trust an app to provide "First-Aid" instructions until you can sync?

 Copy chart

13 responses

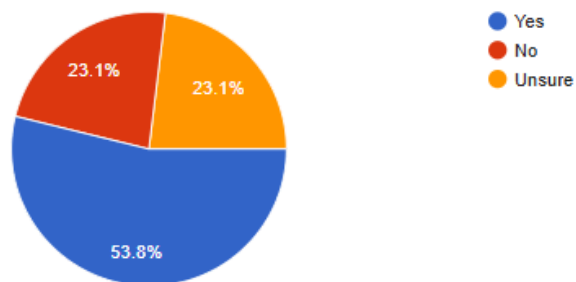


Fig 6: Health Privacy & telemedicine Survey Questionaries 2

Several observations from the patient survey have important implications for the system architecture and user interface.

Respondents' travel time to city hospitals was considerable. Travel times ranged from one to four hours for the vast majority of patients, with some taking between four and eight hours, even full days, and nights to reach urban hospitals, reflecting the spatial layout of rural Nepal described in the literature (Parajuli and Doneys, 2017). This observation supports the need for an application that provides clinical benefit independent of physical clinic visits.

Gender-based constraints on travel were significant among female respondents. Some women participants, especially those between ages 26-60 years old from Rasuwa and Dhading, indicated the need for their family members' permission to seek medical care. Furthermore, the respondents stated that their daily domestic chores often or always precluded their ability to visit clinics for treatment. These findings confirm the gender equity component of the implementation environment outlined in the literature (Parajuli and Doneys, 2017).

There was considerable variation in digital literacy among respondents. Respondents with high educational qualifications (ages 18-25 years) reported smartphone application proficiency of four or five out of five. However, respondents without any formal education and those with primary education (especially older women) reported being comfortable using smartphone applications from one or two out of five. Several respondents without any formal education did not own personal phones. The above heterogeneity demonstrates the conclusion made by Paul et al. (2023) that digital literacy is an important factor in influencing adoption, and further supports NFR5 –

requiring design of the UI to have large touch targets, meaningful iconography, and minimal text entry (SUS score target ≥ 70).

The acceptability of using apps for giving automated guidance for administering first aid when doctors were unavailable owing to poor internet was quite acceptable among smartphone owners. Most smartphone owners said they would rely on the app for providing first-aid guidance when the doctor was not available because of poor internet. Some directly stated yes, and some marked 'unsure.' This demonstrates the clinical acceptability of FR3 – protocol-triggered intervention, and further correlates with the results of the survey among professionals concerning the acceptability of Edge ML.

5.3.2 Clinical Professional Survey Findings

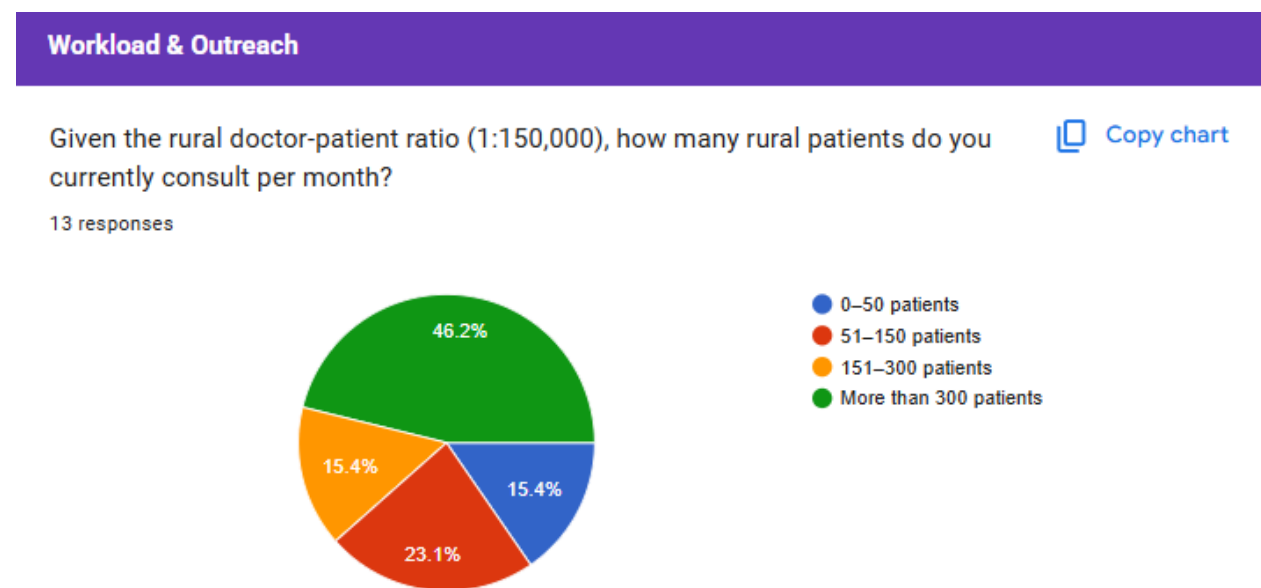


Fig 7: Workload & Outreach Survey Questionnaires

What is the typical delay between a patient recording an anomaly and your ability to review it due to network issues? [Copy chart](#)

13 responses

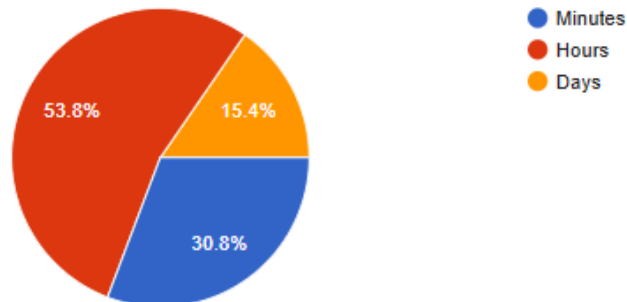


Fig 8: Workload & Outreach Survey Questionaries 2

Technical Constraints

How often does inconsistent data syncing (network jitter) affect your ability to see a coherent trend in a patient's history? [Copy chart](#)

13 responses

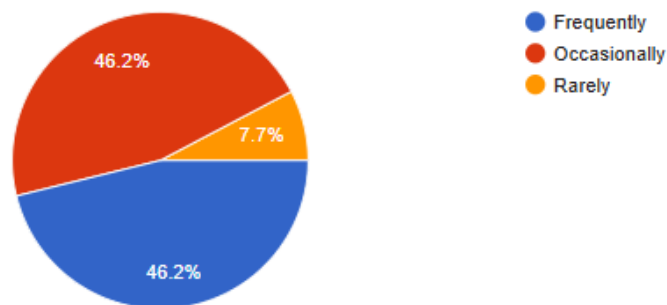


Fig 9: Technical Constraints Survey Questionaries

How confident are you in on-device AI (Edge ML) to trigger alerts for hypertension/diabetes spikes without manual review?

 Copy chart

13 responses

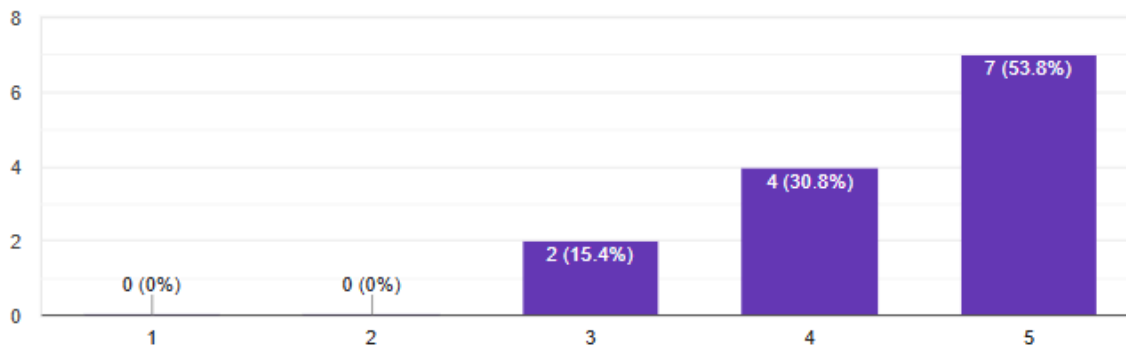


Fig 10: Technical Constraints Survey Questionnaires 2

The information obtained from the clinical professional survey served as strong empirical evidence in support of both the diagnostic assessment of the oversight gap and technical design decisions.

The workload on the patients was substantial; most of the clinicians stated that they worked with more than 300 rural patients monthly. At the observed ratio between the number of rural doctors and patients (1:150,000), the simultaneous and individual monitoring of the patients would be practically impossible without technology. This finding provides empirical evidence for the necessity of implementing an asynchronous overview model via dashboards.

The delay caused by the problems with the Internet connection was stated to be measured in hours or even days by most respondents, while a smaller proportion noted delays lasting just a few minutes. The findings provide evidence for the existence of a significant time gap between the identification of the patient anomalies and their further review by professionals. The FR3 system is designed to assist patients in this very period.

The necessity for monitoring durations of at least six months before significant improvements could be seen in the patients' blood pressure and HbA1c levels received favorable responses in general, with almost all participants giving the importance a value of four or five. This is consistent with what was found by Ngamvitroj, Pitthayapong and Dhippayom (2025) in their study wherein it was discovered that self-management

interventions lasting less than six months had greater decreases in systolic blood pressure, although effectiveness declined in longer durations thereafter.

The effect of network jitter on consistency of data received responses from almost all survey participants as "frequent." Only very few gave the response as "occasionally," while none of the participants said that they experienced it rarely or never in their clinical workflows. These results provide empirical evidence in justifying the jitter synchronization test strategy presented in Section 6.1.2.

The Edge ML alert system was trusted highly among the professionals surveyed, with the majority marking their level of trust at five out of five and most of the rest giving four out of five. Such high levels of trust from professionals can be seen as an endorsement of the scientific validity of the Edge ML system and serve as an appropriate starting point for the validation process of the Edge ML system, where the classification accuracy must match or exceed the approximately 97.5% standard set by Utsha and Morshed (2025).

5.3.3 Cross-Survey Synthesis and Design Implications

Together, the data from the surveys of patients and professionals demonstrate all three compounding factors listed in the problem statement, namely, unreliability of networks, cost-prohibitive nature, and lack of immediate mechanisms to address the issue. Furthermore, the data confirms socio-cultural access barriers, specifically gender-related limitations on mobility that render face-to-face visits to health care providers impossible for a significant share of the target population. All five key research questions stated in Section 4.1 are answered by the data, which directly underpins all functional requirements (FR1-FR8), non-functional requirements (NFR1-NFR6), and criteria for evaluation described in Chapter 5.

Specifically, the digital literacy diversity observed in the patient survey has an immediate implication on the project implementation plan. Namely, the patient-oriented Flutter interface must be developed and tested for use by people who have little to no experience working with smartphones, including individuals with no education at all. This requirement is included in the first-phase implementation task (Section 7.2) as part of the user interface requirements.

Chapter 6: Artefact Planning

This chapter represents the detailed technical specification and architecture of the Swasthya Sathi application. In this regard, this chapter converts the clinical and end-user requirements as identified in previous chapters to tangible functional and non-functional requirements, software/hardware technology stack, and a range of system design artifacts required in the development process during semester two.

6.1 Requirement Analysis

Requirement analysis identifies the essential features and performance standards necessary for Swasthya Sathi to function reliably in low-bandwidth, infrastructure-limited environments. Requirements are categorised into functional and non-functional (how well the system must specifications).

6.1.1 Functional Requirements

The following functional requirements define the core capabilities that the system must deliver. All requirements are treated as essential unless explicitly marked as aspirational.

- FR1 — Offline Data Collection: The Flutter mobile application must enable patients to input blood glucose (mmol/L or mg/dL) and blood pressure (systolic/diastolic mmHg) readings without any internet connectivity. The application must remain fully operable in airplane mode.
- FR2 — On-Device Anomaly Detection: The integrated Edge ML model must process health readings locally and classify each reading into one of three risk categories: No Risk, Moderate Risk, or Critical Risk. Classification must be completed within 5 milliseconds on the target hardware baseline.
- FR3 — Triggered Offline First-Aid Protocols: Upon detection of a Critical Risk classification, the application must automatically retrieve and display the appropriate first-aid protocol from the local FirstAidProtocols table. Protocol display must not require network connectivity and must occur within 2 seconds of the ML classification.
- FR4 — Local Data Persistence: All health readings, risk classifications, ML prediction scores, and protocol activations must be stored persistently in the local SQLite or Hive database, with a sync status boolean accurately reflecting the synchronisation state of each record.
- FR5 — Asynchronous Data Synchronisation: The application must implement a background service that detects available network bandwidth windows and transmits pending records from the SyncQueue to the remote backend server. Synchronisation must be transparent to the user and must not interrupt active application use.
- FR6 — Conflict Resolution: The synchronisation mechanism must implement conflict detection and resolution logic to handle discrepancies between local and server-side

- data, defaulting to a defined conflict resolution policy (e.g., server-wins or timestamp-based last-write-wins, to be determined during implementation).
- FR7 — Doctor Web Dashboard: A web-based interface must enable authenticated medical professionals to review patient health trends, access historical reading data, examine flagged risk alerts with associated ML scores, and record asynchronous notes or medication adjustment recommendations.
 - FR8 — User Authentication: Both the mobile application and the web dashboard must implement secure authentication. Patient and doctor roles must be clearly differentiated, with RBAC enforcing access boundaries.

6.1.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance standards that the system must meet across security, performance, efficiency, and reliability dimensions.

- NFR1 — Data Security: All locally stored patient data must be encrypted using AES-256. The doctor dashboard must enforce Role-Based Access Control, permitting doctors to access only their assigned patients' data. Authentication tokens must be managed securely and refreshed on a defined schedule.
- NFR2 — Performance: Edge ML inference must complete within 5 milliseconds on a device with 2GB RAM and a mid-tier ARM processor. Application launch time must not exceed 3 seconds on the target device baseline.
- NFR3 — Data Efficiency: The synchronisation payload must be minimised through data compression and differential sync strategies. The application must not consume background data in excess of 5MB per week during normal operation.
- NFR4 — Reliability: The system must maintain a data reconciliation success rate of 99% or higher during synchronisation, with no silent data loss. All synchronisation failures must be logged in the SyncQueue with an error message and retry count for diagnostic review.
- NFR5 — Usability: The patient-facing UI must achieve a System Usability Scale (SUS) score of 70 or above in usability testing with representative users. All primary patient workflows must be completable without reading instructional text.
- NFR6 — Cross-Platform Compatibility: The Flutter mobile application must support Android API level 21 and above, targeting the broadest feasible range of devices available in the Nepali rural market.

6.2 Software Requirements

The software stack is selected to prioritise cross-platform compatibility, offline data resilience, and computational efficiency on resource-constrained hardware.

Component	Technology	Justification
Mobile Framework	Flutter (Dart)	Single codebase for Android and iOS; strong offline-first library ecosystem; native performance for ML inference.
Local Database	SQLite / Hive	SQLite for relational data integrity; Hive for Flutter-native NoSQL performance. Both support encrypted storage.
Web Dashboard	React / Next.js	Component-based architecture; server-side rendering for performance; broad ecosystem support for data visualisation.
Edge ML Runtime	TensorFlow Lite	Optimised for mobile inference; supports model quantisation to reduce size and latency on constrained hardware.
ML Models	Logistic Regression / Decision Tree	Computationally lightweight; clinically validated for binary and multi-class risk classification; interpretable outputs.
Backend API	Node.js / Express or Firebase	Lightweight REST API for synchronisation endpoints; Firebase offers offline-capable real-time database as alternative.
Security	AES-256, JWT, RBAC	AES-256 for local storage encryption; JWT for stateless authentication; RBAC middleware on all protected endpoints.

6.3 Hardware Requirements

6.3.1 Development Environment

Development requires a workstation with a minimum of 8GB RAM (16GB recommended) to support concurrent Flutter compilation, ML model training, and web development tooling. A minimum of 20GB free storage is required for SDK installations, emulator images, and project assets.

6.3.2 Deployment Environment

Patient-end devices must be Android smartphones capable of running API level 21 or above, with a minimum of 2GB RAM to support the Flutter application and TensorFlow Lite inference. The device must have local storage sufficient to maintain the SQLite/Hive database across extended offline periods. Doctor-end access is provided through any web-capable device (desktop, laptop, or tablet) with a modern browser, requiring no dedicated hardware installation.

6.4 Feasibility Study

The feasibility of the Swasthya Sathi platform is assessed across five key dimensions, with an overall positive outlook for its implementation in rural Nepal.

Technical Feasibility

The project's technical foundation is rated HIGH, utilizing mature, production-validated technologies like the Flutter SDK for mobile and React/Next.js for the doctor dashboard. Implementing Edge ML on resource-constrained devices is technically viable, as established benchmarks show that lightweight models can achieve inference latencies of under 5ms even on devices with only 2GB of RAM. While the asynchronous synchronization mechanism is the most complex component due to the unstable network conditions in rural Nepal, it is supported by established mathematical frameworks for conflict resolution.

Operational Feasibility

Operational feasibility is rated MEDIUM-HIGH, supported by survey data indicating that patients are willing to trust automated first-aid guidance when doctors are unavailable. A primary operational risk is the digital literacy gap among older rural populations; however, this is addressed through a UI design that avoids text-heavy workflows and prioritizes ease of use. On the provider side, doctors have expressed high confidence in the system, noting that the asynchronous model fits their existing workflows for managing large rural patient panels.

Financial Feasibility

This dimension is rated HIGH because the project utilizes free, open-source development tools such as TensorFlow Lite, SQLite, and the Flutter SDK, resulting in zero development costs for the academic prototype. Furthermore, the architecture is designed for long-term commercial sustainability; the offline-first approach keeps data costs extremely low (targeting less than 5MB

per week per patient), and on-device ML inference eliminates the need for expensive cloud computing resources.

Legal and Ethical Feasibility

The project faces a MEDIUM feasibility rating in this area due to a regulatory vacuum in Nepal regarding telemedicine malpractice and data privacy laws. To mitigate ethical risks, the platform uses only WHO-approved clinical protocols and is explicitly positioned as a first-response guidance tool rather than a diagnostic instrument. Data security is managed through industry-standard AES-256 encryption and Role-Based Access Control (RBAC) to ensure patient privacy and prevent unauthorized access.

Scheduling Feasibility

The 13-week implementation timeline is rated MEDIUM and is considered achievable but tight for a solo developer. The most significant schedule risk is the development of the custom synchronization module. To prevent timeline slippage, the project employs a modular dependency structure and maintains a strict scope boundary, focusing exclusively on diabetes and hypertension to prevent feature drift.

6.5 System Design

6.5.1 Use Case Diagram

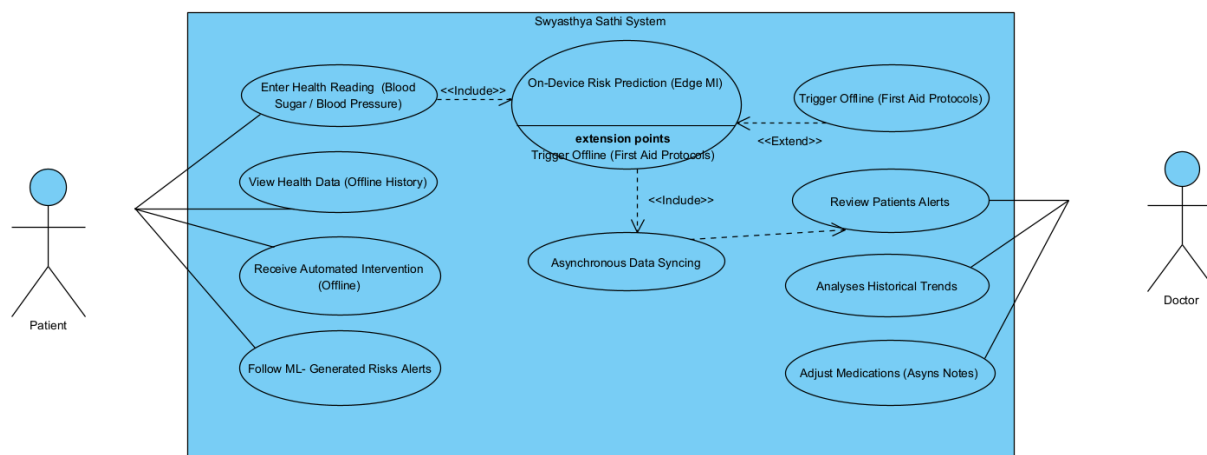


Fig 11: Use Case Diagram of Sywasthya sathi

The use case diagram of the Healthcare Management System specifies the unique interaction patterns for the two main actors – the Patient (from rural Nepal) and the Doctor (from remote/urban areas) – regarding their interaction with the system's functional components. The Patient actor interacts with the offline first layer of the system and engages in activities such as providing health readings, checking local data, getting automatic interventions, and ML-based notifications. On the other hand, the Doctor actor is limited to interacting with the system through the web interface and focuses on activities such as reading alerts, analysis of historical

trends, and prescription adjustments. The boundaries of the system include the mobile application and the web interface, with an asynchronous synchronization layer that becomes active with low bandwidth window detection acting as the connection between the two layers.

6.5.2 Database Schema

The database schema is structured into six interrelated tables that together manage the complete offline-first workflow, from initial data entry through to professional review and synchronisation reconciliation.

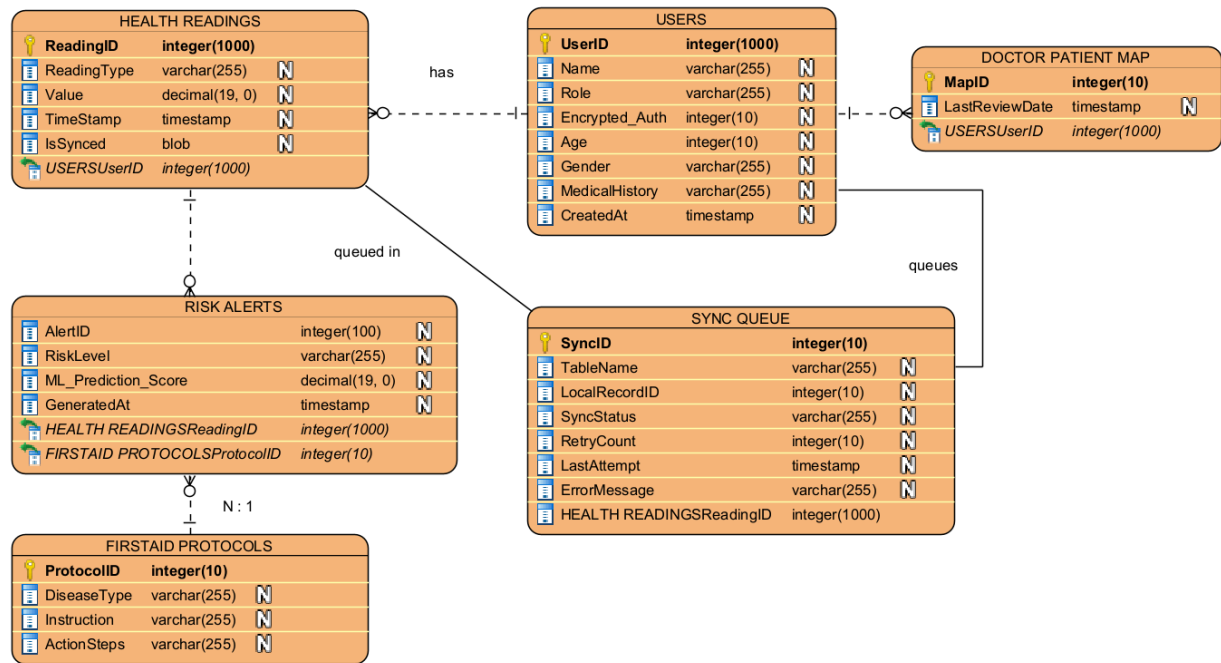


Fig 12: Entity Relationship Diagram of Sywasthya sathi

1. Users Table: Stores UserID (UUID, Primary Key), Name (VARCHAR), Role (VARCHAR: Patient or Doctor), Encrypted_Auth (TEXT), Age (INT), Gender (VARCHAR), MedicalHistory (TEXT), and CreatedAt (TIMESTAMP). This table serves as the authentication and identity foundation for both patient and doctor actors.
2. HealthReadings Table: Captures ReadingID (UUID, Primary Key), UserID (Foreign Key to Users), ReadingType (VARCHAR: Blood_Sugar or Blood_Pressure), Value (DECIMAL), Timestamp (TIMESTAMP), and IsSynced (BOOLEAN). The IsSynced flag is the critical field for managing the offline-to-online synchronisation lifecycle of each record.

3. RiskAlerts Table: Stores AlertID (UUID, Primary Key), ReadingID (Foreign Key to HealthReadings), RiskLevel (VARCHAR: Critical, Moderate, or No Risk), ML_Prediction_Score (DECIMAL), ProtocolID (Foreign Key to FirstAidProtocols), and GeneratedAt (TIMESTAMP). This table directly links clinical readings to ML outputs and intervention protocols.
4. FirstAidProtocols Table: Contains ProtocolID (SERIAL, Primary Key), DiseaseType (VARCHAR), Instructions (TEXT), and ActionSteps (TEXT). All entries are sourced from WHO-approved or Nepal-specific clinical guidelines and are stored entirely on-device, enabling offline retrieval.
5. DoctorPatientMap Table: Manages MapID (SERIAL, Primary Key), DoctorID (Foreign Key to Users), PatientID (Foreign Key to Users), and LastReviewDate (TIMESTAMP). This relational table enforces the RBAC constraint that doctors can only access data for patients explicitly assigned to them.
6. SyncQueue Table: Handles SyncID (SERIAL, Primary Key), TableName (VARCHAR), LocalRecordID (UUID), SyncStatus (VARCHAR: Pending, Synced, or Conflict), RetryCount (INT), LastAttempt (TIMESTAMP), and ErrorMessage (TEXT). This table is the operational core of the conflict resolution and resilient syncing mechanism.

6.5.3 Activity Diagram

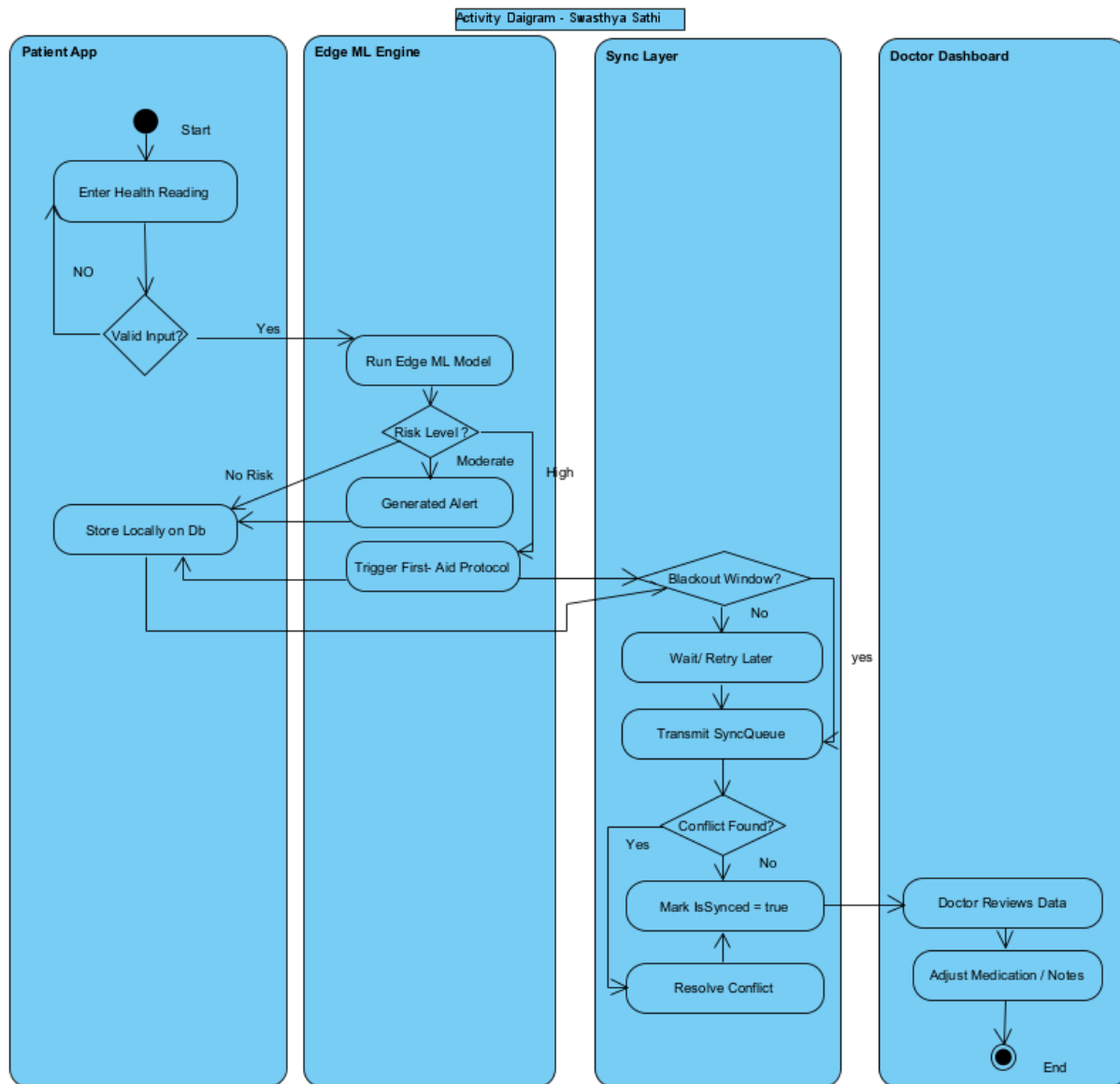


Fig 13: Activity Diagram of Sywasthya sathi

The system activity diagram illustrates the sequential and conditional logic flow through which a patient interaction with the mobile application is processed from data entry to professional review. The flow initiates with the patient entering a health reading, which triggers immediate on-device ML analysis. The ML model's output determines the subsequent branch: if no critical risk is detected, the reading is stored locally and queued for synchronisation. If a critical risk is detected, the protocol trigger fires before local storage, ensuring that the patient receives immediate guidance without any delay. The synchronisation branch is conditionally activated by bandwidth window detection and operates asynchronously with respect to the patient's active use of the application. The professional review branch is activated by the doctor's access to the web dashboard and operates entirely within the synchronised, online context of the RBAC-protected dashboard.

Chapter 7: Critical Analysis and Evaluation

This chapter details the comprehensive strategy for verifying the technical integrity and clinical efficacy of Swasthya Sathi. Evaluation is organised across two parallel tracks: a testing strategy focused on technical validation of system components, and an evaluation strategy focused on clinical outcomes and user acceptability. Together, these tracks address the full spectrum of quality criteria relevant to a system with both engineering and healthcare dimensions.

7.1 Testing Strategy and Plan

The testing strategy is designed to validate the system's performance under the specific technical constraints identified through the literature review and primary research: high network jitter, limited mobile hardware resources, variable bandwidth availability, and the imperative for data integrity across asynchronous synchronisation cycles.

7.1.1 Edge ML Validation

Edge ML testing will evaluate two primary dimensions: predictive accuracy and inference latency. Following benchmarks established in the clinical ML literature, the project targets a classification accuracy of approximately 97.5% across all three risk categories (No Risk, Moderate Risk, Critical Risk) for both blood glucose and blood pressure readings. This target is deliberately grounded in subject-wise cross-validation rather than random-split validation, addressing the methodological concern identified in the literature review that random-split metrics may significantly overestimate real-world clinical performance.

Inference latency testing will verify that the TensorFlow Lite model completes its classification within the 5-millisecond target on a device representative of the lower end of the target hardware spectrum (2GB RAM, mid-tier ARM processor). Unit tests will verify that the model correctly triggers Critical Risk alerts for clinically defined threshold values: blood glucose above 10 mmol/L (hyperglycaemia threshold) and systolic blood pressure above 180 mmHg (hypertensive crisis threshold), among others drawn from WHO clinical guidelines.

Test Cases — Edge ML Validation:

TC	Test Case	Input	Expected Output	Acceptance Criterion	Method
TC-M-L-01	Critical hyperglycaemia detection	Blood glucose reading: 11.5 mmol/L (above 10 mmol/L WHO threshold)	Classification: Critical Risk; ML_Prediction_Score > 0.85; RiskAlerts record created; FirstAidProtocols trigger fired	100% detection on all Critical threshold inputs	Unit test (automated)
TC-M-L-02	Hypertensive crisis detection	Systolic BP: 185 mmHg / Diastolic BP: 115 mmHg	Classification: Critical Risk; alert and protocol triggered within 5ms inference	100% detection; latency < 5ms	Unit test + stopwatch profiling

TC-M L-0 3	Moderate risk classification	Blood glucose: 7.5 mmol/L (elevated but below critical threshold)	Classification: Moderate Risk; alert generated; no first-aid protocol triggered; reading stored with IsSynced = false	Correct moderate-class output; no false Critical trigger	Unit test (automated)
TC-M L-0 4	No-risk classification	Blood glucose: 5.2 mmol/L; Systolic BP: 118 mmHg — both within normal range	Classification: No Risk; no alert generated; reading stored locally; IsSynced = false	Zero false positive Critical alerts on normal-range inputs	Boundary value analysis (automated)
TC-M L-0 5	Subject-wise cross-validation accuracy	Full clinical dataset split by patient ID (subject-wise, not random); 80/20 train/test split	Weighted F1-score ≥ 0.975 ; sensitivity for Critical class ≥ 0.97 ; specificity ≥ 0.96	$\geq 97.5\%$ accuracy — benchmark from Utsha & Morshed (2025)	Scikit-learn cross_val_score + subject-aware KFold

7.1.2 Synchronisation and Reconciliation Testing

To mitigate Risk R1 (Data Sync Conflicts), the synchronisation mechanism will undergo a comprehensive suite of 'Jitter Testing' as scheduled in the Semester 2 Gantt chart. This testing regime involves simulating a range of adverse network conditions — including variable bandwidth, high latency, packet loss, and mid-transfer connectivity interruptions — to verify that the asynchronous syncing mechanism correctly reconciles offline local data with the doctor's dashboard without data loss, record duplication, or silent corruption.

Specific test scenarios will include: (a) simultaneous offline data entry and active synchronisation to test concurrent write handling; (b) deliberate introduction of conflicting records in the local and remote databases to verify conflict resolution logic; (c) simulation of extended offline periods (48–72 hours) followed by bulk synchronisation to assess performance under high-volume sync conditions; and (d) partial connectivity interruption during synchronisation to verify that the SyncQueue correctly records the failure state and resumes without data loss.

Test Cases — Synchronisation and Reconciliation:

TC#	Test Case	Simulated Condition	Expected Behaviour	Pass Criterion	Tool
TC-S Y-01	Clean sync — stable connection	100Mbps stable WiFi; 50 pending HealthReadings records in SyncQueue with SyncStatus = Pending	All 50 records transmitted; IsSynced set to true; SyncStatus = Synced; doctor dashboard reflects all 50 readings within 10 seconds	100% records synced; zero data loss; dashboard updated	Manual + Postman

TC-S Y-02	High-jitter mobile network sync	3G simulation: 100ms average latency, ± 80 ms jitter, 5% packet loss via tc netem	Sync completes without data corruption; RetryCount incremented on transient failures; all records eventually reach SyncStatus = Synced; no silent failures	$\geq 99\%$ of records synced within 3 retry cycles; zero silent data loss	adb shell tc netem + Postman
TC-S Y-03	Mid-transfer connectivity drop	WiFi disconnected mid-sync after 30% of SyncQueue transmitted; connectivity restored after 60 seconds	System detects interruption; partially transmitted records retain SyncStatus = Pending with error logged in ErrorMessage; sync resumes from last successful checkpoint on reconnection; no duplication on server	Zero duplicate records on server; all pending records eventually synced; no data corruption	Manual WiFi toggle + database inspection
TC-S Y-04	Conflict resolution — last-write-wins	Same ReadingID modified locally (Timestamp T1) and on server (Timestamp T2 > T1) during offline	Timestamp-based last-write-wins policy applied; server version (T2) preserved; local version overwritten; conflict logged in	Correct version preserved per policy; conflict record	Database direct manipulation + API inspection

		period; triggered	sync	SyncQueue SyncStatus Conflict resolved	with = then	logged; no data lost	
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7.1.3 Security and Protocol Testing

Security testing will encompass two distinct sub-components. First, penetration testing will be conducted against the local database encryption implementation to verify the integrity of AES-256 protection for stored patient data, and against the dashboard's RBAC implementation to confirm that cross-patient data access is effectively prevented. Second, protocol trigger testing will verify that the offline first-aid protocol system correctly retrieves and displays the appropriate WHO-approved instructions for each possible risk classification, under zero-connectivity conditions, within the 2-second display requirement defined in FR3.

TC#	Test Case	Attack / Test Scenario	Expected Defence	Pass Criterion	Tool
TC-S E-01	Local database extraction	adb pull of SQLite database file from patient device; direct inspection of file contents with SQLite browser	All stored data (HealthReadings, RiskAlerts, Users) appears as encrypted binary; no plaintext patient data readable from raw file	Zero plaintext patient data recoverable from extracted database file	adb pull + SQLite Browser

TC-S E-02	Cross-patient data access (RBAC breach attempt)	Authenticated doctor attempts to access patient data outside their assigned DoctorPatientMap via direct API call with modified patient ID parameter	API returns HTTP 403 Forbidden; no patient data returned; access attempt logged in security audit log	100% of cross-patient access attempts blocked; all logged	Postman / curl API testing
TC-S E-03	JWT token forgery	Crafted JWT token with forged doctor role claim and invalid signature; presented to protected dashboard endpoint	Token validation fails at signature check; API returns HTTP 401 Unauthorised; forged token rejected without exposing any patient data	Zero successful authentications with forged tokens	jwt.io token crafter + Postman
TC-S E-04	Critical Risk first-aid protocol — display time	Device in airplane mode; blood glucose reading of 12 mmol/L entered; ML classifies as Critical Risk	Correct WHO-approved diabetes protocol retrieved from local FirstAidProtocols table and displayed on screen within 2 seconds of classification; no network request made	Protocol displayed within 2 seconds at zero connectivity; correct protocol for disease type	Stopwatch + airplane mode + network inspector

7.2 Evaluation Strategy and Plan (DECIDE Framework)

The following evaluation plan for the Swasthya Sathi platform is structured according to the DECIDE framework, drawing on the clinical, technical, and socio-cultural requirements established in the sources.

D – Determine the Goals

The primary goal is to evaluate if Swasthya Sathi effectively bridges the "oversight gap" in rural Nepal by providing clinically reliable and technically correct chronic disease monitoring. Specific objectives include:

Achieving measurable improvements in metabolic risk factor control (blood glucose and blood pressure).

Ensuring the system is usable for patients with limited digital literacy (target SUS score ≥ 70).

Reducing gender-based barriers to healthcare access.

E – Explore the Questions

The evaluation seeks to answer several critical questions:

Does persistent, offline-first monitoring lead to better health outcomes compared to existing episodic consultation models?

Can the platform's automated first-aid protocols effectively reduce patient vulnerability when a doctor is unavailable?

Does the platform provide a private, non-judgmental environment that reduces a female user's dependence on supervised travel?

Can users across the digital literacy spectrum navigate the interface without instructional mediation?

C – Choose the Evaluation Methods

A multi-method approach will be utilized:

Clinical/Technical: Subject-wise cross-validation for Edge ML to ensure generalizability to unseen patients. Automated latency profiling and adversarial network simulation for synchronization integrity.

Usability: A structured survey employing the System Usability Scale (SUS) for quantitative feedback.

Socio-Cultural: Qualitative interviews and retention metrics to assess long-term adoption and the reduction of access barriers.

I – Identify the Practical Issues

Several constraints must be managed during the evaluation:

The Digital Divide: Addressing the low connectivity rates (as low as 14% in Karnali) and the prevalence of unstable 3G networks.

Data Access: The need for validated, anonymized Nepali clinical datasets to ensure the Edge ML models reflect the local population's metabolic characteristics.

Timeline: Executing a comprehensive validation (including jitter testing and security audits) within the final four weeks of the 13-week implementation schedule.

D – Decide How to Deal with Ethical Issues

To navigate the current "regulatory vacuum" in Nepal's telemedicine laws, the following measures are in place:

Safety: All first-aid protocols are exclusively WHO-approved and must be reviewed by a medical professional.

Privacy: Implementation of AES-256 encryption for local storage and Role-Based Access Control (RBAC) to prevent unauthorized data access.

Informed Consent: Ensuring all research participants provide consent and that data is handled anonymously.

Positioning: Explicitly defining the platform as a first-response guidance tool, not a diagnostic instrument.

E – Evaluate, Analyze, Interpret, and Present the Data

Quantitative Analysis: Analyzing ML F1-scores (target ≥ 0.975), mean inference latency (target $< 5\text{ms}$), and synchronization success rates (target $\geq 99\%$).

Qualitative Analysis: Interpreting user feedback on trust and adoption through the lens of human-element frameworks.

Reporting: Presenting findings through structured documentation, including the final project validation reports scheduled for late 2026.

Chapter 8: Implementation Plan

This chapter provides a phased, technically detailed roadmap for the implementation of Swasthya Sathi during Semester 2. The plan translates the requirements and design specifications established in Chapter 5 into a structured sequence of development activities, mapped directly to the milestones defined in the Semester 2 Gantt chart. The implementation strategy prioritises local-first functionality, ensuring that the system's core offline capabilities are operational and tested before the integration of network-dependent features.

8.1 Implementation Strategy Overview

The process will be based on a modular and phased approach divided into five phases in total. These phases are created such that each one depends only on the deliverables of the previous phase and does not have any dependency on anything that has not been implemented and tested yet. First, the mobile application and intelligence will be created and implemented, followed by integration with the remote doctor's dashboard. Synchronization between the two applications will only be attempted once both the parts are individually functional.

This is done keeping in mind the architecture chosen for our design where the network independent functions form the minimum requirements and everything else is additional.

8.2 Phase 1: Environment Setup and UI Scaffolding

This initial phase establishes the technical foundation by configuring the Flutter SDK for the mobile app and React/Next.js for the doctor dashboard. Key activities include initializing the SQLite/Hive local database and developing the primary user interfaces for both patients and clinicians, focusing on low cognitive load for users with limited digital literacy.

8.3 Phase 2: Local Intelligence and Security Integration

Phase 2 operationalizes the system's offline capabilities by developing and optimizing Edge ML models (Logistic Regression and Decision Tree) for risk classification. These models are converted to TensorFlow Lite for high-speed, on-device inference. Simultaneously, AES-256 encryption is implemented for local data persistence, and a Role-Based Access Control (RBAC) framework is established to ensure data privacy.

8.4 Phase 3: Intervention Logic and Remote Oversight

This phase completes the clinical guidance system by integrating WHO-approved first-aid protocols that trigger automatically based on ML risk outputs, even without internet. The doctor dashboard is finalized with trend visualizations and an asynchronous notes interface. The phase concludes with integration testing to verify the end-to-end offline workflow from data entry to protocol display.

8.5 Phase 4: Integration and Resilient Syncing

Phase 4 addresses the technical challenge of inconsistent networks by introducing an asynchronous synchronization layer. This includes a background service for bandwidth detection and conflict resolution logic that uses a "last-write-wins" policy, while flagging critical risk conflicts for manual professional review. Preliminary jitter testing is conducted here to resolve synchronization weaknesses under simulated adverse conditions.

8.6 Phase 5: Validation, Testing, and Final Delivery

The final phase involves a comprehensive evaluation, including formal jitter testing to ensure a 99% data reconciliation rate and a security audit of the encryption and RBAC systems. A usability survey is conducted to measure the System Usability Scale (SUS) score. The project culminates in the compilation of technical documentation and a final demonstration of the complete end-to-end workflow

Chapter 9: Conclusion

9.1 Summary of Findings

With respect to the context in which Swasthya Sathi is to operate, this contextual report has built the foundations required for the design and development of a robust offline-first telemedicine platform that can withstand and thrive in the low bandwidth conditions found in rural Nepal. From the research presented in Chapters 3 and 4, it is evident that the current telemedicine models have failed for the very simple reason that the problem lies in their architecture, not conceptually.

The proposed architecture will overcome these shortcomings by implementing a suite of architectural designs that include local-first data storage with SQLite/Hive, local machine learning model for clinical assessment with Edge ML (TensorFlow Lite, Logistic Regression/Decision Tree), automatically triggered response based on protocols with locally available WHO approved first aid protocol library, and asynchronous synchronization with doctor dashboard. With regards to risks involved in developing this project, there are a total of five risks identified, all with High impact.

9.2 Project Impact

Swasthya Sathi has the potential to deliver meaningful impact across three dimensions. Clinically, by providing persistent, offline-capable monitoring for patients managing diabetes and hypertension, it directly addresses the oversight gap that leaves approximately 80% of rural

Nepal's population without reliable professional guidance. The automated first-aid protocol system ensures that patients receive actionable clinical guidance during the inevitable delays between anomaly detection and professional review — a capability that has no equivalent in current market offerings.

The socio-cultural aspect of the platform's impact on healthcare delivery is linked to the issue of overcoming gender constraints that exist within the current healthcare delivery mechanism. In this regard, telemedicine can serve as an important instrument for increasing gender equality by offering users a means of managing their personal health without the need to travel and with anonymity.

On an architectural level, the platform introduces a replicable approach for offline-first patient monitoring that can be useful beyond the confines of Nepal. The modularity of this approach allows the use of the offline layer in isolation from the other two – the synchronization and dashboard layers.

9.3 Future Recommendations

Several directions for future development merit consideration beyond the scope of this project. First, while the Edge ML models provide high classification accuracy in controlled testing, continuous subject-wise validation with real-world clinical data from Nepali patient populations is essential to maintain clinical defensibility over time. A feedback loop in which doctor-reviewed outcomes are used to retrain and refine the models would significantly strengthen the system's long-term clinical reliability.

Secondly, digital literacy support can be improved by the development of a community health workers' training program, where trusted local intermediaries can help facilitate patient adoption and sustained usage. Research consistently highlights this component as a key factor in the sustainable implementation of digital health programs.

Thirdly, in future versions, the clinical application of the technology can be expanded to cover more common chronic illnesses, including COPD and malnutrition tracking in rural Nepal. Additionally, inexpensive wearable sensors can be incorporated into the platform, minimizing patient effort in recording clinical parameters manually

9.4 Final Conclusion

Swasthya Sathi represents a rigorous and contextually grounded application of software engineering principles to one of the most consequential public health challenges in Nepal. By

prioritising local intelligence over cloud dependency, asynchronous communication over synchronous video, and clinical utility over feature richness, the system is designed to succeed precisely where conventional telemedicine has failed. The research, planning, and design work documented in this contextual report provides a solid and comprehensive foundation for the Semester 2 implementation phase. If delivered as specified, Swasthya Sathi will demonstrate that software engineering, when disciplined by clinical evidence and shaped by the specific constraints of its deployment context, can play a transformative role in achieving healthcare equity for geographically disadvantaged populations.

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Appendix

Project Proposal

Part1 - Project Proposal

Student Name	Aayush Man Shakya
Student Number	2337692
Degree Pathway	BSc(Hons) Software Engineering
Supervisor Name	Pawan KC
Title of Project	Swasthya Sathi - "Offline-First" Telemedicine System: Bridging the Healthcare Gap in Rural Nepal
Abstract of the project	This project involves the research, design, and development of an offline-first telemedicine platform specifically engineered for the extreme low-bandwidth environments of rural Nepal. Currently, 80% of the population is underserved due to a critical doctor shortage (ratios as low as 1:150,000) and the failure of traditional, video-dependent telehealth in areas with poor connectivity. This system bridges the healthcare gap by utilizing Edge-based ML and local data storage to provide continuous monitoring without a constant internet connection. A key innovation is the integration of offline first-aid precautions, which provide immediate medical guidance to patients during the unavoidable delays in asynchronous doctor reviews caused by connectivity and professional shortages.
Project deliverables	1. Project Artifact 2. Final Year Report 3. Project Poster 4. Source Code Repository 5. Project Documentation 6. Testing and evaluation Report 7. Project Demonstration
Description of your artefact	Background Nepal's healthcare system is characterized by a severe disparity between urban and rural areas. While the national doctor-to-patient average is 1:3,000, remote locations suffer with ratios as low as 1:150,000. Approximately 80% of the population remains underserved due to poor infrastructure, low salaries for medical staff, and a significant "brain drain" of professionals moving abroad. Existing telehealth solutions are often unusable in these regions because they rely on high-bandwidth, constant internet connections which are inconsistent in rural Nepal.

	Problem Statement Traditional healthcare and existing telehealth solutions are ineffective in rural Nepal because they rely on high-bandwidth, constant internet connections that are often unavailable. High data costs for video consultations and the severe "brain drain" of medical professionals create a dual barrier: patients cannot connect to doctors, and even when they do, doctors are stretched too thin to provide immediate responses. This leaves patients with chronic conditions, such as diabetes and hypertension, in a dangerous "oversight gap" where they have no access to professional guidance or immediate intervention tools when health anomalies occur in offline settings Proposed Solution The proposed Swasthya Sathi - "Offline-First" Health Monitoring System is designed to prioritize functionality in low-bandwidth environments through the following strategies: • Local-First Architecture: Utilizing on-device storage (SQLite/Hive) and Edge ML to ensure health monitoring and anomaly detection continue even with zero connectivity. • Asynchronous Communication: A tiered strategy that favors low-data text-based consultation and background data syncing over high-bandwidth video, which is treated only as an optional enhancement. • Immediate Offline Intervention: To mitigate the delay in doctor reviews, the system includes a locally stored library of first-aid protocols. These verified medical precautions trigger automatically upon the detection of an anomaly, providing life-saving guidance entirely offline while the system waits for a low-bandwidth window to sync data with a professional. Aim • To research, design, and develop a secure, functional, and offline-capable telemedicine platform featuring on-device Machine Learning, specifically tailored to bridge the patient-doctor gap in rural Nepal and improve health outcomes for chronic disease management. Objectives: • To investigate and analyze offline-first architectures and edge computing for healthcare applications. • To develop a cross-platform mobile application using the Flutter framework that supports local database storage and offline functionality. • To implement Edge ML models (such as Logistic Regression or Decision Trees) for real-time, on-device anomaly detection and risk prediction. • To engineer an asynchronous syncing mechanism that transmits critical health alerts and summaries to a doctor's dashboard when network permits. • To design a triggered first-aid protocol system that provides immediate, localized medical instructions to patients during the "waiting period" for professional review.
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	List of Features: • Patient Mobile App: Features local data collection for diabetes (blood sugar) and hypertension (blood pressure), automatic anomaly detection, and low data consumption. • Edge ML Integration: On-device risk prediction and early warning alerts to reduce dependency on cloud processing. • Doctor Web Dashboard: Provides comprehensive patient overviews, trend analysis of historical health patterns, and alert notifications for critical conditions. • Tiered Communication: A strategy favoring asynchronous text messages for low bandwidth, with voice and video calls as secondary options for urgent cases. • Offline First-Aid Protocols: Locally stored, text-based medical instructions triggered by Edge ML alerts to provide immediate stabilization steps for chronic risks.					
Risk Analysis	Risk ID	Possible Risks	Probability	Impact	Likelihood	Approach to Mitigate
	R1	Data Sync Conflicts	medium	High	Medium	Implement robust database reconciliation and conflict resolution logic for offline-to-online syncing.
	R2	Edge ML Inaccuracy	medium	High	Medium	Use established algorithms like Logistic Regression and perform rigorous testing with local datasets to ensure reliable alerts.
	R3	Data Breaches	low	High	Medium	Implement encryption for local databases and Role-Based Access Control (RBAC) for the doctor dashboard.

	R4	Time Management	medium	High	High	Create a detailed project timeline with specific weekly milestones for both app and dashboard development.
	R5	Incorrect First-Aid Guidance	Low	High	Low	Use only standardized, WHO-approved or Nepal-specific clinical guidelines for the offline precaution library.
How does your project relate to your degree course?	This project integrates complex software engineering skills acquired during my studies: • Mobile & Web Development: Utilizing Flutter for cross-platform mobile development and web technologies for the doctor dashboard. • Machine Learning: Implementing on-device ML analysis for health insights, moving beyond basic cloud-based AI. • Database Management: managing local databases (like SQLite or Hive) for offline storage and handling data synchronization. • Networking: Researching and applying asynchronous communication protocols to handle low-bandwidth constraints.					
Resources required in developing the artefact	1. Software and Technologies: Flutter for mobile, a web framework (like React or Next.js) for the dashboard, and local database tools. 2. Hardware: A development computer with a minimum of 8 GB RAM. 3. Other: Access to medical datasets for ML training and academic journals regarding rural healthcare in Nepal.					
Have you completed & submitted your ethics form?	YES			NO		
	If the project is a development of previous work by yourself or others, give details below. Failing to declare such previous work here may be treated as an academic offence					

References

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<https://english.onlinekhabar.com/doctors-reluctance-serve-remote-nepal.html>
<https://www.sciencedirect.com/science/article/abs/pii/S0895435618310813>
<https://trueconf.com/blog/productivity/low-bandwidth-communication>
<https://sqlite.org/docs.html>
<https://dl.acm.org/doi/10.1145/3359591.3359737>

Supervisor Signature:


2026-2-27

After the proposal has been signed off by both the supervisor and course coordinator scan the proposal and upload on BREO with signatures. Projects that follow proposals that have not been approved may be cancelled and there will be no compensation for any time lost

Part 2 – List of relevant resources

Fill in this section after your project proposal has been approved by your supervisor. Use Harvard referencing (see <https://lrweb.beds.ac.uk/a-guide-to-referencing>). Modify the list below as appropriate. This list is part of Assignment 1 and will be submitted with the Project Proposal.

1. Books

- a. Guijarro, D. (2024) *Scalable Application Development with NestJS*. Birmingham: Packt Publishing.
- b. Magolan, G., Housley, P., de Peretti, A., Bell, J. and Guijarro, D. (2019) *Nest.js: A Progressive Node.js Framework: Hit the ground running with Nest.js*. Birmingham: Packt Publishing.

2. Journal Papers

- a. K. Singh et al., "Design and implementation of telemedicine network in a sub Himalayan State of India," HEALTHCOM 2006 8th International Conference on e-Health Networking, Applications and Services, New Delhi, India, 2006, pp. 78-83, doi: 10.1109/HEALTH.2006.246423.

- b. G. Maiga and I. Arinaitwe, "A model for telemedicine adoption in remote healthcare settings," 2017 IST-Africa Week Conference (IST-Africa), Windhoek, Namibia, 2017, pp. 1-8, doi: 10.23919/ISTAFRICA.2017.8102405.
- c. M. J. Chang, J. K. Jung, M. W. Park and T. M. Chung, "Strategy to reinforce security in telemedicine services," 2015 17th International Conference on Advanced Communication Technology (ICACT), PyeongChang, Korea (South), 2015, pp. 170-175, doi: 10.1109/ICACT.2015.7224778.

3. Web Sites with relevant information

- a. <https://english.onlinekhabar.com/doctors-reluctance-serve-remote-nepal.html>
- b. <https://www.tandfonline.com/doi/full/10.1080/16549716.2024.2385177?scroll=top&needAccess=true#abstract>

4. Relevant software

- a. React.js and flutter as a framework for frontend development and node.js as a framework for backend development.
- b. Sqlite to interact with database

5. Relevant hardware

- a. A development computer/laptop with a minimum of 8 GB Ram.

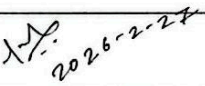
6. Other

- a. Access to academic journals, framework documents for node.js and tutorial videos on realtime will be helpful for the development of my project.

Ethics Form

FACULTY OF CREATIVE ARTS, TECHNOLOGIES AND SCIENCE

Form for Research Ethics Projects (Ethics Form)

Student Name	Aayush Man Shakya
Student Number	2337692
Degree Pathway	BSc(Hons) Software Engineering
Supervisor name	Pawan KC
Supervisor Signature	 2026-2-28
Title of project	Swasthya Sathi - "Offline-First" Telemedicine System: Bridging the Healthcare Gap in Rural Nepal

SECTION A

Please summarise the ethical issues involved and how they will be addressed.

In any proposal involving human participants please provide information on how:

- informed consent will be obtained
- confidentiality will be observed
- the nature of the research and the means of dissemination of the outcomes will be communicated to participants.

SECTION B Check List

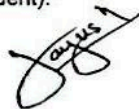
Please answer the following questions by circling YES or NO as appropriate.

Does the study involve vulnerable participants or those unable to give informed consent (e.g. children, people with learning disabilities, your own students)?	YES
	☒ NO
Will the study require permission of a gatekeeper for access to participants (e.g. schools, self-help groups, residential homes)?	YES
	☒ NO
Will it be necessary for participants to be involved without consent (e.g. covert observation in non-public places)?	YES
	☒ NO
Will the study involve sensitive topics (e.g. obtaining information about sexual activity, substance abuse)?	YES
	☒ NO
Will blood, tissue samples or any other substances be taken from participants?	YES
	☒ NO
Will the research involve intrusive interventions (e.g. the administration of drugs, hypnosis, physical exercise)?	YES
	☒ NO
Will financial or other inducements be offered to participants (except reasonable expenses or small tokens of appreciation)?	YES

	☒ NO
Will the research investigate any aspect of illegal activity (e.g. drugs, crime, underage alcohol consumption or sexual activity)?	YES
	☒ NO
Will participants be stressed beyond what is considered normal for them?	YES
	☒ NO
Will the study involve participants from the NHS (patients or staff) or will data be obtained from NHS premises?	YES
	☒ NO

If the answer to any of the questions above is "Yes", or if there are any other significant ethical issues, then further ethical consideration is required. Please document carefully how these issues will be addressed.

Signed (student):



Date: 27/02/2026

Countersigned (Supervisor):


2026-2-27

Date: 27/02/2026

Weekly Report

UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR RMET PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Aayush Man Shakya	Supervisor's Name: Pawan KC
Project Title: Swasthya Sathi : "Offline-First" Telemedicine System: Bridging the Healthcare Gap in Rural Nepal	
Week No: 6	Report No: 1

Summary of progress (including any problems)	• Topic Refinement: Refined the project focus on telemedicine solutions for rural Nepal with limited bandwidth, ensuring the research problem clearly addresses connectivity challenges in remote healthcare delivery. • Preliminary Research: Conducted initial exploration of existing telemedicine platforms and remote healthcare models used in low-resource environments. This helped identify key technological and operational gaps related to unstable internet connectivity and data transmission. • System Concept Development: Started conceptualizing a low-bandwidth telemedicine architecture, focusing on offline-first design principles where medical data can be stored locally and synchronized when network connectivity becomes available.
Plan for next week	• Begin drafting the technical framework for the telemedicine system architecture. • Explore machine learning approaches suitable for edge devices and low computational environments. Identify relevant academic papers on telemedicine implementation in developing countries. • Continue collecting sources for the literature review section.

Supervisor's comments	
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Student's Signature
Date 16th March, 2026

Supervisor's Signature
Date 2026-3-16

UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR RMET PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Aayush Man Shakya	Supervisor's Name: Pawan KC
Project Title: Swasthya Sathi : "Offline-First" Telemedicine System: Bridging the Healthcare Gap in Rural Nepal	
Week No: 7	Report No: 2

Summary of progress (including any problems)	Technical Framework: Defined the tech stack, choosing Flutter for cross-platform development and SQLite/Hive for local-first data persistence Objective Setting: Established clear research goals, including investigating Edge ML (Logistic Regression/Decision Trees) for on-device anomaly detection and engineering asynchronous syncing mechanisms Risk Assessment: Completed the initial Risk Analysis (R1-R5), identifying high-impact risks such as Data Sync Conflicts and Edge ML Inaccuracy, with associated mitigation strategies Ethics Clearance: Confirmed and submitted the required Ethics Form for research involving medical data and human subjects
Plan for next week	- Initiate the Literature Review focusing on the clinical efficacy of telemedicine for diabetes and hypertension. - Research the Bashshur evaluation framework to align research inputs and outputs with global standards. - Investigate Available Bandwidth (ABW) estimation challenges in low-resource networks.
Supervisor's comments	

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Student's Signature *Aayush*
Date .. *16th March, 2026*

Supervisor's Signature *1.1*
Date *2026-3-16*

UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR RMET PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Aayush Man Shakya	Supervisor's Name: Pawan KC
Project Title: Swasthya Sathi : "Offline-First" Telemedicine System: Bridging the Healthcare Gap in Rural Nepal	
Week No: 8	Report No: 3

Summary of progress (including any problems)	Literature Review Completion: Successfully completed an extensive review of the sources, establishing the clinical efficacy of telemedicine in managing chronic conditions like Type 2 Diabetes (HbA1c reduction) and hypertension (systolic/diastolic BP reduction) Socio-Cultural Analysis: Integrated research regarding rural Nepal, confirming that telemedicine effectively reduces gender-based barriers and improves time management for household chores by providing localized access to care Architecture Design: Finalized the "Offline-First" framework, selecting Flutter for cross-platform development and SQLite/Hive for secure local data storage to ensure functionality during zero-connectivity periods
Plan for next week	Primary Research Instrument Design: Develop a structured questionnaire and interview guide for rural Nepalese participants, drawing on the "acceptability" and "accessibility" dimensions identified in research on telemedicine in rural Nepal. The study will specifically focus on identifying socio-cultural barriers such as gender-based travel restrictions and time management for household chores. Clinical Data Benchmarking: Conduct a comparative study of clinical data from existing telemedicine trials for Type 2 Diabetes (T2D) and hypertension. I will use the HbA1c reduction targets (-1.5%) and systolic/diastolic blood pressure benchmarks found in the sources to establish "success" metrics for the Swasthya Sathi research model.

	• Baseline Data for "Waiting Period" Guidance: Research and compile baseline data on average asynchronous delay times in existing rural health networks. This data will be used to justify the research into immediate offline intervention protocols (first-aid guidelines) for the "oversight gap" when professional reviews are delayed.
Supervisor's comments	* Complete the L-R and start working in the data survey.

Student's Signature *Aayush*
Date 16th March, 2026

Supervisor's Signature *[Signature]*
Date 2026-3-16

UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR RMET PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Aayush Man Shakya	Supervisor's Name: Pawan KC
Project Title: Swasthya Sathi: "Offline-First" Telemedicine System: Bridging the Healthcare Gap in Rural Nepal	
Week No: 9	Report No: 4

Summary of progress (including any problems)	Field Data Survey Execution: Successfully conducted the Healthcare Access & Telemedicine Acceptability Study, gathering 26 total responses from rural patients and healthcare practitioners in Nepal Validation of the "Oversight Gap": Survey data confirmed that 53.8% of doctors consult over 300 patients per month, while 61.5% report delays of hours or even days in reviewing health anomalies due to poor connectivity High Clinical Trust in Edge ML: Achieved strong validation for autonomous alerts, with 84.6% of practitioners rating their confidence in on-device AI as a 4 or 5 out of 5 Patient Accessibility & Trust: Identified that patients often face day-long or overnight travel to reach hospitals and expressed high trust in using a mobile app for "Offline First-Aid" instructions during internet outages Technical Literature Synthesis: Integrated survey findings with research on GAN-enhanced hybrid deep learning and model-driven architectures to refine the system's low-latency diagnostic capabilities
Plan for next week	Consolidate research and objectives to build a structural outline, then draft the core conceptual framework and supporting sections before finalizing the logical flow and visual diagrams.
Supervisor's comments	* Work on the System Architecture

Student's Signature *[Signature]*
Date 30th March, 2026

Supervisor's Signature *[Signature]*
Date 2026-3-28

Healthcare Access & Telemedicine Acceptability Study – Swasthya Sathi (Responses)

Form_Responses										
1	Timestamp	Age	Gender	District/Region	Education Level	On average, how long does it take you to react	Do you require permission from a family member	How often do household responsibilities (fair)	Do you own a personal mobile phone?	How com
2	3/20/2026 8:26:48	18-25	Male	Sindhupalchok	Secondary	1-4 hours		4 Often	Yes	
3	3/20/2026 8:27:49	60+	Female	Sindhupalchok	No formal education	1-4 hours		5 Always	No	
4	3/20/2026 8:28:39	41-60	Male	Dhading	Primary	4-8 hours		1 Sometimes	Yes	
5	3/20/2026 8:29:45	41-60	Female	Rasuwa	Primary	4-8 hours		4 Often	Yes	
6	3/20/2026 8:30:24	26-40	Female	Rasuwa	Secondary	4-8 hours		4 Always	Yes	
7	3/20/2026 8:31:04	26-40	Female	Rasuwa	No formal education	4-8 hours		4 Often	No	
8	3/20/2026 8:31:52	26-40	Male	Dhading	Primary	1-4 hours		1 Sometimes	Yes	
9	3/20/2026 8:32:47	26-40	Female	Tanahu	Primary	Day-long/Overnight		4 Sometimes	Yes	
10	3/20/2026 8:33:26	18-25	Female	Rasuwa	Higher Education	1-4 hours		3 Often	Yes	
11	3/20/2026 8:34:03	18-25	Female	Dhading	Higher Education	1-4 hours		4 Rarely	Yes	
12	3/20/2026 8:34:45	18-25	Male	Solukhumbu	Secondary	Day-long/Overnight		5 Sometimes	Yes	
13	3/20/2026 8:36:18	41-60	Female	Dhading	No formal education	1-4 hours		4 Rarely	No	
14	3/20/2026 8:37:06	41-60	Male	Rasuwa	Secondary	Day-long/Overnight		5 Always	Yes	
15										
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Clinical Efficacy & Technical Constraints in Rural Telemedicine

Form_Responses						
Timestamp	Given the rural doctor-patient ratio (1:150,000)	What is the typical delay between a patient re	In your clinical experience, is a 6-month moni	How critical is remote monitoring (via MMAS	How often does inconsistent data syncing (n	How confident are you in on-device AI (Edge I
3/20/2026 8:14:58	51-150 patients	Minutes		4	5 Occasionally	5
3/20/2026 8:15:50	151-300 patients	Hours		5	5 Rarely	5
3/20/2026 8:16:13	More than 300 patients	Hours		5	3 Frequently	5
3/20/2026 8:16:36	More than 300 patients	Hours		5	4 Occasionally	4
3/20/2026 8:17:01	More than 300 patients	Hours		3	3 Frequently	3
3/20/2026 8:17:26	More than 300 patients	Hours		5	5 Frequently	5
3/20/2026 8:17:54	51-150 patients	Days		3	2 Frequently	5
3/20/2026 8:18:21	More than 300 patients	Days		4	5 Occasionally	5
3/20/2026 8:18:52	More than 300 patients	Hours		4	1 Frequently	4
3/20/2026 8:19:15	0-50 patients	Minutes		4	3 Occasionally	5
3/20/2026 8:19:37	151-300 patients	Hours		5	4 Occasionally	3
3/20/2026 8:19:59	0-50 patients	Minutes		2	2 Occasionally	4
3/20/2026 8:20:21	51-150 patients	Minutes		4	3 Frequently	4